Spectral–Spatial–Temporal MAP-Based Sub-Pixel Mapping for Land-Cover Change Detection

Da He, *Member, IEEE*, Yanfei Zhong, *Senior Member, IEEE*, and Liangpei Zhang, *Fellow, IEEE*

***Abstract*— The maximum *a posteriori* (MAP) estimation model-based sub-pixel mapping (SPM) method is an alternative way to solve the ill-posed SPM problem. The MAP estimation model has been proven to be an effective SPM approach and has been extensively developed over the past few years, as a result of its effective regularization capability that comes from the spatial regularization model. However, various spatial regularization models do not always truly reflect the detailed spatial distribution in a real situation, and the over-smoothing effect of the spatial regularization model always tends to efface the detailed structural information. In this article, under the scenario of time-series observation by remote sensing imagery, the joint spectral–spatial–temporal MAP-based (SST_MAP) model for SPM is proposed. In SST_MAP, a newly developed temporal regularization model is added to the MAP model, based on the prerequisite for a temporally close fine image covering the same study region. This available fine image can provide the specific spatial structures most closely conforming to the ground truth for a more precise constraint, thereby reducing the over-smoothing effect. Furthermore, the three dimensions are mutually balanced and mutually constrained, to reach an equilibrium point and achieve restoration of both smooth areas for the homogeneous land-cover classes and a detailed structure for the heterogeneous land-cover classes. Four experiments were designed to validate the proposed SST_MAP: three synthetic-image experiments and one real-image experiment. The restoration results confirm the superiority of the proposed SST_MAP model. Notably, under the background of time-series observation, SST_MAP provides an alternative way of land-cover change detection (LCCD), achieving both detailed spatial-scale and high-frequency temporal LCCD observation for the study case of urbanization analysis within the city of Wuhan in China.**

***Index Terms*— Joint spectral–spatial–temporal (SST) model, land-cover change detection (LCCD), maximum *a posteriori* (MAP) estimation model, spatial regularization model, sub-pixel mapping (SPM), temporal regularization model.**

I. Introduction

THE mixed-pixel problem is a ubiquitous phenomenon in remotely sensed imagery, which means that one sampled pixel consists of multiple land-cover classes on the Earth's surface [1], due to the limitation of the frequency of the instantaneous field of view (IFOV) of the sensor. This phenomenon has seriously impeded the development of land-use/land-cover (LULC) information extraction [2], since image classification is implemented pixel-by-pixel, and one pixel should be assigned only one label of a land-cover class. When encountering a mixed pixel, the one dominant land-cover class within the mixed pixel can conquer the whole mixed pixel and efface the other possible content. This means that the mixed pixel can be mistakenly identified as a pure pixel, which is counter-intuitive and will compromise the classification result.

To this end, soft classification, which is also called "spectral unmixing," has long been an effective solution for the mixed-pixel problem. This approach attempts to decompose the mixed pixel into a weighted combination of several pure land-cover classes, based on a pre-constructed linear or non-linear mixture model [3]–[6]. In this way, one mixed pixel is proportionally assigned to multiple classes, instead of only one label. The output of spectral unmixing is a set of abundance images displaying the proportion of each class within each mixed pixel. However, the spatial variation frequency of the output abundance images still remains in the original scale, and a more detailed spatial structure cannot be provided.

The sub-pixel mapping (SPM) technique was developed to characterize the specific spatial distribution of each component within a mixed pixel, as an extension of spectral unmixing, to improve the frequency of the spatial variation compared to the original coarse image. SPM is generally conducted by segmenting a mixed pixel into several sub-pixels and assigning each sub-pixel with class attributes. The number of sub-pixels assigned to each class is guided by the abundance images and the spatial allocation of each class is determined by the spatial dependence assumption [7], [8]. The spatial dependence assumption describes a universal principle derived from geographical phenomena, i.e., observations of a given property that resembles each other are inclined to be spatially close to each other [9], [10]. In this way, SPM precisely specifies the location of each pure land-cover class within the mixed pixels and improves the frequency of the spatial variation of the original coarse image.

Multiple SPM methods have been developed over the past decades. The early-stage SPM algorithms were focused on spatial dependence optimization. Such algorithms include the spatial attraction model-based SPM (SASM) [11]; the pixel swapping algorithm-based SPM (PSSM) [12]; neural network

Manuscript received September 20, 2018; revised June 2, 2019; accepted September 8, 2019. Date of publication December 18, 2019; date of current version February 26, 2020. This work was supported by the National Natural Science Foundation of China under Grant 41622107 and Grant 41771385. *(Corresponding author: Yanfei Zhong.)*

The authors are with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan 430079, China, and also with the Hubei Provincial Engineering Research Center of Natural Resources Remote Sensing Monitoring, Wuhan University, Wuhan 430079, China (e-mail: heda@whu.edu.cn; zhongyanfei@whu.edu.cn; zlp62@whu.edu.cn).

Color versions of one or more of the figures in this article are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TGRS.2019.2947708

0196-2892 © 2019 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:58:24 UTC from IEEE Xplore. Restrictions apply.

models such as Hopfield neural networks (HNNs), [13] and back-propagation sub-pixel mapping (BPSM) [10]; and evolutionary computation models [14] such as genetic algorithm sub-pixel mapping (GASM) [15], differential evolution sub-pixel mapping (DESM) [16], clonal selection sub-pixel mapping (CSSM) [17], particle swarm optimization (PSO) [18], and multiagent systems [19]. However, the spatial dependence assumption is only fulfilled on the condition that the intrinsic scale of the spatial variation in each land-cover class is not smaller than the sampling scale imposed by the image pixels [17]. When encountering detailed structural restoration of a certain land-cover class that is finer than the sampling scale, the spatial dependence assumption seems to be invalid. This being so, a more specific spatial distribution constraint is required.

Subsequently, a series of geostatistic-based SPM methods have been proposed, using a geostatistic model to simulate the spatial patterns that are smaller than the sampling scale. These methods include geometric methods [20], vectorial boundary methods [21], indicator kriging methods [22], [23], and the multiple-point simulation model [24]. Alternative methods such as Markov random field (MRF)-based methods [25] and conditional random field (CRF)-based methods [26] have also been derived for the same purpose. However, these model-derived spatial patterns can only simulate for a regional variation and limited kind of pattern, which may not characterize all land cover types in a research region.

In fact, SPM is an ill-posed problem since it attempts to restore a small amount of information (possessed by the coarse image) to a larger amount of information (possessed by the fine image), amounting to an increase in the information entropy, which is impossible without extra complimentary information. The abundance image can only provide proportional information retained on the coarse spatial scale and extra finer-scale spatial distribution information should be incorporated to increase the information entropy and regularize the ill-posed problem [27]. However, deficient corroborative spatial patterns to characterize the real geographical distribution will mislead the final restoration. Meanwhile, uncertainty inherently exists in the process of spectral unmixing and will propagate to the subsequent SPM process if the abundance values are directly accepted for SPM objective construction, as in the majority of the SPM methods.‘

Aiming at the regularization of the SPM problem, the maximum *a posteriori* (MAP) model has been proposed. The MAP model estimates the unobserved spatial distribution of sub-pixels on the basis of the observed abundance images and the spatial regularization model. Notably, it is the MAP estimation rule and the spatial regularization model that constrain the ill-posed SPM problem to be a well-posed problem. Furthermore, the spatial regularization model is able to balance the uncertainty derived from the abundance images, which overcomes the uncertainty propagation problem in the traditional manner of separating unmixing and SPM [28]. Last but not least, the MAP model has the advantages of expansibility, which permits the extension of the regularization model for various demands [28], as different regularization models can characterize a different spatial distribution of sub-pixels, according to the specific requirement. Many different methods have been developed based on the MAP model, including total variation (TV)-based MAP [28], [29], non-local-based MAP [30], multishifted image-based MAP [31], [27], as well as sparse representation-based MAP [32], which are all able to improve the reconstruction results. However, the spatial regularization model has limitations. It can take the smoothness and continuity into consideration, but it cannot provide heterogeneous distribution information, such as discretely distributed buildings, regardless of the fact that certain land-cover classes have their own spatial distribution patterns, which cannot be simulated by only one spatial regularization model. For this reason, the spatial regularization model is only suitable for homogeneous land-cover classes and is unsuitable for heterogeneous land-cover classes, such as those found in urban areas with intricate distributions. As noted in [28], the MAP-based methods may cause over-smoothing. Thus, a more appropriate spatial constraint is urgently needed for heterogeneous land-cover restoration.

With the extensive development of Earth observation by remote sensing imagery, time-series observations of specific research areas are now becoming available for multiple satellite sensors. This being so, a temporally close fine image (TCFI) can be incorporated to help with the restoration of the coarse images in the time-series as it can provide the specific spatial patterns or structures for a more precise constraint. A TCFI generally refers to an image acquired by a different satellite sensor with a finer spatial resolution at a different acquisition time. Multiple spatio-temporal sub-pixel mapping (SSM) algorithms have been developed to incorporate a TCFI in the SPM process. For example, a TCFI can be used to construct a land-cover change (LCC) rule as prior information for constraining the sub-pixel allocation [33], [34], or a TCFI can be used to construct a spatio-temporal dependence model, to iteratively search for the optimal configuration of the subpixel location [35], [36]. However, the incorporation of a TCFI is sub-pixel-wise processing (i.e., taking only one pixel at a time), the mechanism of which neglects the spatial continuity within land-cover objects and has a tendency to fall into local optima. Furthermore, this mechanism of considering only the local similarity of each sub-pixel cannot learn an integrated spatial representation for a faithful constraint of the SPM problem.

In this article, based on this prerequisite, the spectral–spatial–temporal MAP-based (SST_MAP) model is proposed. Benefiting from the expansibility of the MAP model, the TCFI can be formulated as a temporal regularization term and incorporated into the MAP model. The three dimensions (spectral, spatial, and temporal) are simultaneously considered to formulate the joint SST_MAP model. The spectral term constrains the proportion of each land-cover class; the spatial term preserves the smoothness and continuity of the land covers and balances the uncertainty of the abundance maps; and the temporal term provides the extra specific fine spatial pattern constraint for preserving the detailed spatial structure and neutralizes the over-smoothing effect of the spatial term. In this way, the three dimensions are mutually balanced and mutually constrained to reach an equilibrium point and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:58:24 UTC from IEEE Xplore. Restrictions apply.

achieve the restoration of both smooth areas for homogeneous land-cover classes and a detailed structure for heterogeneous land-cover classes. Moreover, the MAP framework is able to model the SPM problem for the entire images (both the TCFI and abundance images) in each optimization iteration, instead of the traditional pixel-wise or sub-pixel-wise optimization. This approach is more computationally effective and tends to reach the global optimal solution, traversing more integrated spatial representations for further regularization.

Furthermore, the proposed SST_MAP model can be directly applied to the prevalent applications of LCC detection (LCCD). LCCD is a technique of using time-series Earth observation imagery to detect and analyze the land surface variation of a given area [37]. LCCD is of great importance for understanding the global change in the vegetation cover, since it plays an important role in the decision-making of human activities, such as ecosystem management [38], cultivation planning [39], urbanization [40], etc. However, due to the limited spatio-temporal resolutions inherent in satellite imagery-based Earth observation, LCCD products may not always capture ground-based changes [41]. Specifically, the Landsat Thematic Mapper (TM) classification product cannot completely match the growth periodicity of some crops, because the phase from green-up to maturity is a dynamic and rapid process, i.e., a 10–12-day growth period, which cannot be detected and captured by the 16-day Landsat TM products [42]. On the other hand, high temporal resolution products with frequent, dense observations (e.g., daily Moderate Resolution Imaging Spectroradiometer (MODIS) land-cover products) are usually confronted with the coarse spatial resolution and mixed-pixel problem, which seriously impede the extraction of fine-scale change information [43]. In this case, the proposed SST_MAP model can be adopted to improve the spatio-temporal resolution of the time-series satellite observations. SST_MAP first restores the coarse images, with the TCFI serving as the spatial constraint. The restored thematic map can then be compared to the TCFI to yield the change map. In this way, the SST_MAP model can achieve both the fine-spatial scale and high-frequency temporal LCCD observations.

The main contributions and innovations of the proposed SST_MAP model can be summarized as follows.

1) The SST_MAP model for SPM is proposed. In SST_MAP, the newly developed temporal regularization model can fully exploit the spatio-temporal information from the TCFI, incorporating the specific spatial distribution patterns from the TCFI to further regularize the ill-posed SPM problem. It can restore and preserve the detailed spatial structure, thus neutralizing the over-smoothing effect of the spatial term. This incorporation is based on the expansibility of the MAP model, in that the TCFI can be easily incorporated into the joint processing and simultaneous optimization.
2) In SST_MAP, due to the incorporation of the temporal regularization model, the three dimensions of remote sensing imagery are all taken into consideration. The interaction of these three dimensions during the iteration optimization converges the solution to reach an equilibrium point and achieves restoration of both the smooth areas for homogeneous land-cover classes and the detailed structure for heterogeneous land-cover classes.
3) With respect to the LCCD application in Experiment 4, the proposed SST_MAP model was adopted to restore the coarse images within the time-series observation, and the restoration result was then matched to the TCFI for urbanization-related LCCD, achieving the purpose of both the fine spatial resolution and fine temporal resolution LCCD maps, which could be further expanded to the observation series.

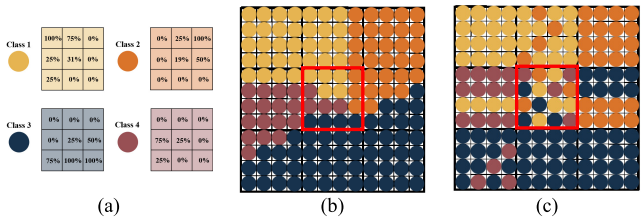


Fig. 1. Specific example of SPM processing. (a) 3 × 3 grid of mixed pixels, with the abundance values for the four classes. (b) Each pixel is divided into 16 sub-pixels (4 × 4) with a scale factor of 4, and assigned with endmember class labels based on the abundance value. (c) Another possible spatial configuration for the four classes.

The rest of this article is organized as follows. Section II provides a background to SPM problem modeling. Section III describes the proposed SST_MAP algorithm in detail, from modeling to solution. The experiments implemented to validate the proposed algorithm are described in Section IV, followed by the sensitivity analysis of the critical parameters in Section V. Finally, a discussion and conclusion are provided in Section VI.

II. Background

The SPM technique is able to specify the spatial distribution of each component within a mixed pixel, as an extension of spectral unmixing, to improve the frequency of the spatial variation compared to the original coarse image. It is generally conducted by first segmenting a mixed pixel into several smaller units of sub-pixels, with the specific number of generated sub-pixels determined by the pre-defined scale factor S. Each sub-pixel is then allocated to a certain land-cover class attributes. This allocation is based on two principles: 1) the number of sub-pixels assigned to each class is guided by the proportion value within the abundance images (generally called the abundance constraint) and 2) the spatial configuration of each class is determined by the spatial dependence assumption. The spatial dependence assumption describes a universal principle derived from geographical phenomena, i.e., observations of a given property that resemble each other are inclined to be spatially close to each other [9], [10].

Fig. 1 provides a specific comprehensible example of the SPM operation, in which four land-cover classes are included. A 3 × 3 grid of input abundance images of each class is displayed in Fig. 1(a), with the corresponding proportion values inside each grid (a grid generally refers

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:58:24 UTC from IEEE Xplore. Restrictions apply.

to a mixed pixel). According to the abundance sum-to-one constraint (ASC) in the spectral unmixing process [6], the sum of four classes inside each grid is equal to 100%. Each coarse pixel is then segmented by a scale factor of 4, i.e., one coarse pixel is divided into 16 (4×4) sub-pixels, as shown in Fig. 1(b) and (c). Following the segmentation, each sub-pixel is allocated to a certain class, based on the proportion value, for which we refer the interested reader to [49]. After the class attribute allocation, the sub-pixel scale thematic map can be generated, as shown in Fig. 1(b) and (c). However, due to the ill-posed property of SPM, multiple solutions can be reached from only one set of abundance images, e.g., Fig. 1(b) and (c) are two possible configurations which are both subject to the abundance constraint. Therefore, abundance information alone cannot constrain the problem and extra information is needed to regularize the SPM result to a unique solution. Generally speaking, spatial dependence assumption is one of the most classical constraints for regularizing sub-pixels' allocation, based on which the configuration where the sub-pixels of the same class are closely connected [as shown in Fig. 1(b)] is more likely to conform to the real land-cover distribution.

Numerous models have been proposed to provide a sufficient constraint for sub-pixel spatial allocation, as reviewed in the Introduction. In this article, the spatial regularization model and the proposed temporal regularization model are jointly applied in the MAP estimation framework, to simultaneously constrain the ill-posed SPM problem.

III. SST_MAP SPM Algorithm

A. SPM Observation Model Construction

In this article, the MAP estimation framework is adopted, to regularize the SPM problem. It can achieve an optimal spatial distribution of sub-pixels on the basis of the observed abundance images and the spatial regularization model. Furthermore, based on the advantages of expansibility for further adding the regularization model, a newly developed temporal regularization model is proposed and incorporated into the MAP estimation framework, which can fully exploit the spatio-temporal information from the time-series images to further regularize the ill-posed SPM problem. Specifically, the spatio-temporal relationship between the TCFI at time T_1 and the solution high-resolution thematic image (HRTI) at T_2 was constructed as the temporal regularization model, to help with the restoration of the coarse image at T_2.

Step 1 *(Downsampling Observation Model):*

In the first place, an observation model is needed to simulate the downsampling process from HRTI to its low-resolution abundance images (both at T_2), to quantitatively define a downsampling model between two different observation scales.

The downsampling observation model can be depicted as shown in Fig. 2, where y^1 is the abundance image of class 1 and x^1 is the high-resolution classification for class 1. That is, if the sub-pixel belongs to class 1, it is attributed to 1; otherwise, it is attributed to 0. The coordinate system is applied to x and y, respectively, as shown in Fig. 2, with each pixel and sub-pixel annotated by a specific coordinate. For instance,

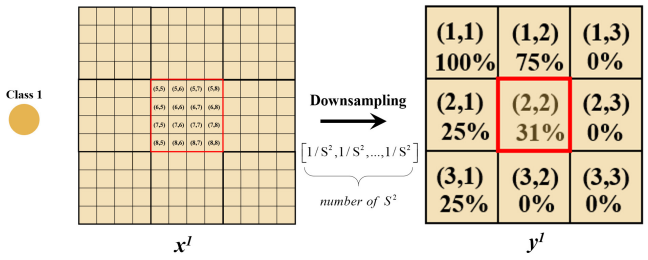


Fig. 2. Simulated downsampling process from the solution HRTI x^1 to the low-resolution abundance images y^1 for class 1.

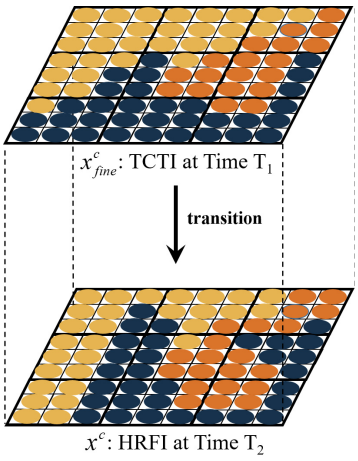


Fig. 3. Transition process from the TCFI x^c_{fine} to the solution HRTI x^c for class c.

the pixel $y(2,2)$ corresponds to the footprint from sub-pixels $x(5,5), x(5,6), \ldots, x(8,8)$. The downsampling scale is four in this case, and the downsampling matrix is simplified to an averaging filter, in which all the elements are equal to $1/S^2$, in this case, $1/16$. We take the center pixel of y and its corresponding footprint in x as an example, as highlighted by the red square, where five ($31\% \times 16$) sub-pixels belong to class 1, and thus five of the sub-pixels within the sub-pixels $x(5,5)$ to $x(8,8)$ are equal to 1, while the others are equal to 0. An equation characterizing the numerical relationship between $y(2,2)$ and its corresponding footprint in x can be comprehensively represented as

$$y^1(2,2) = \underbrace{[1/16, 1/16, \ldots, 1/16]}_{\text{number of } 16} * [x^1(5,5), x^1(5,6), \ldots, x^1(8,8)]^T. \quad (1)$$

By extending this mapping relationship between the abundance images and the HRTI, the downsampling observation model can be established. For each land-cover class c, the downsampling observation model is assumed as follows:

$$y^c = Dx^c + n_d^c \quad (2)$$

where D is the downsampling matrix, which consists of multiple inverse scale factors $1/S^2$ as elements. y^c and x^c are the abundance images and the HRTI for class c, respectively, and n_d^c is the additive noise.

Step 2 *(Transition Observation Model):*

In time-series observation, the TCFI x^c_{fine} has a strong spatial correlation with the solution HRTI x^c, and is able to provide

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:58:24 UTC from IEEE Xplore. Restrictions apply.

the sub-pixel scale spatial pattern specific to the research area. In order to make full use of the TCFI x_{fine}^c, a temporal regularization model is proposed to characterize the spatio-temporal relationship between the TCFI x_{fine}^c and the solution HRTI x^c, which we called the "transition observation model." The establishment of the transition observation model is based on the idea of simulating the transition process from TCFI x_{fine}^c at time T_1 to the solution HRTI x^c at time T_2 (as shown in Fig. 3), which can be denoted as

$$x^c = T x_{\text{fine}}^c + n_t^c \tag{3}$$

where T is the transformation matrix at the sub-pixel scale for class c and n_t^c is the additive noise for class c.

The transformation matrix is extremely difficult to simulate and extra training data are needed, which can be difficult to obtain [36]. Instead, in the proposed model, the acquisition of the transformation matrix is based on the temporal dependence assumption [36]. Originating from the phenomenon of analogous geographic elements having a tendency to be closer than disparate ones, analogical reasoning can be made, so that spatial correlation also exists between temporal images and temporally close observations of a given property are more inclined to assemble together than temporally distant observations. Since the available TCFI x_{fine}^c is temporally close to the HRTI x^c, most of the land-cover classes remain unchanged from x_{fine}^c to x^c. This being so, x_{fine}^c can be seen as a benchmark of the solution x^c and the matrix T can be assumed to be an identity matrix. Moreover, the main purpose of the transition observation model is to bridge the spatial relationship of the bitemporal images and make use of the fine spatial patterns for reconstruction, and not to simulate the transition process. Thus, it is feasible to adopt this simplification strategy.

It is worth noting that the less the changes occurred during the transformation process, the more similar the TCFI and HRTI are, and more spatial patterns can be used for the restoration of HRTI. Based on this idea, the constructed temporal regularization model can serve as a discriminator to automatically select the unchanged spatial patterns for a more precise HRTI restoration. Therefore, the time interval and change variation during the transformation process are critical to the usability of the TCFI; a large time interval and dramatic change variation can make the TCFI useless. The impact of the time interval and change variation is analyzed in Section IV.

B. MAP-Based SPM

The MAP estimation model is adopted for its capability of incorporating spectral information (specifically, the abundance images), temporal information (specifically, the TCFI), and spatial prior information, while simultaneously considering the impact of these factors for HRTI x^c estimation. The MAP objective can be formulated as the *posteriori* probability of x^c in the presence of y^c and x_{fine}^c, and the MAP approach searches for the most plausible x^c (the final solution is annotated as $\widehat{x}_{\text{MAP}}^c$, for distinction) to maximize the *posteriori* probability, which can be described as

$$\widehat{x}_{\text{MAP}}^c = \arg\max \left\{ \Pr(x^c | y^c, x_{\text{fine}}^c) \right\}. \tag{4}$$

Based on Bayes' rule, the *a posteriori* probability can be transformed into the multiplication of several probability density functions

$$\widehat{x}_{\text{MAP}}^c = \arg\max \left\{ \frac{\Pr(y^c, x_{\text{fine}}^c | x^c) \Pr(x^c)}{\Pr(y^c)} \right\} \tag{5}$$

where y^c is known in advance, thus $\Pr(y^c)$ is a constant and can be removed. Since y^c and x_{fine}^c are mutually exclusive, $\Pr(y^c, x_{\text{fine}}^c | x^c)$ can be split by the multiplication form of the two probabilities

$$\widehat{x}_{\text{MAP}}^c = \arg\max \left\{ \Pr(y^c | x^c) \Pr\left(x_{\text{fine}}^c | x^c\right) \cdot \Pr(x^c) \right\}. \tag{6}$$

The logarithm function is then applied to (6) on both sides to convert the multiplication form to a summation form, which does not change the monotonicity of the equation

$$\widehat{x}_{\text{MAP}}^c = \arg\max \left\{ \log \Pr(y^c | x^c) + \log \Pr\left(x_{\text{fine}}^c | x^c\right) + \log \Pr(x^c) \right\} \tag{7}$$

where $\Pr(y^c | x^c)$ is the probability density of y^c in the presence of x^c, $\Pr(x_{\text{fine}}^c | x^c)$ is the probability density of x_{fine}^c in the presence of x^c, and $\Pr(x^c)$ is the *a priori* probability of x^c.

The probability density $\Pr(y^c | x^c)$ models the error in the downsampling observation model (2), which is assumed to be Gaussian-distributed, with zero mean and the variance σ [44], and the probability density can be given by

$$\Pr(y^c | x^c) = \prod_{\forall x, y} \frac{1}{\sigma \sqrt{2\pi}} \exp\left(-\frac{(y^c - D x^c)^2}{2\sigma^2} \right) \tag{8}$$

In the same way, the probability density $\Pr(x_{\text{fine}}^c | x^c)$ models the error in the transition observation model (3), which is also assumed to be Gaussian-distributed, and can be given by

$$\Pr\left(x_{\text{fine}}^c | x^c\right) = \prod_{\forall x_{\text{fine}}, x} \frac{1}{\sigma' \sqrt{2\pi}} \exp\left(-\frac{\left(x_{\text{fine}}^c - x^c\right)^2}{2\sigma'^2} \right). \tag{9}$$

The spatial prior term $\Pr(x^c)$ can be represented by the Gibbs energy function as

$$\Pr(x^c) = \frac{1}{\rho} \exp\left(-\frac{1}{\beta} U(x^c) \right) \tag{10}$$

where ρ is a constant for normalization, β is a control parameter, and $U(x^c)$ is the energy of the spatial distribution of class c at the target spatial resolution. Substituting (8)–(10) into (7) can generate

$$\widehat{x}_{\text{MAP}}^c = \arg\max \left\{ \begin{array}{l} 2 \log \dfrac{\left(\sigma \sqrt{2\pi}\right)^{-N}}{\rho} - \sum \dfrac{(y^c - D x^c)^2}{2\sigma^2} \\ - \sum \dfrac{\left(x_{\text{fine}}^c - x^c\right)^2}{2\sigma'^2} - \dfrac{1}{\beta} U\left(x^c\right) \end{array} \right\}. \tag{11}$$

This equation can be simplified by eliminating trivial constants and taking the negative symbol on both sides of the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:58:24 UTC from IEEE Xplore. Restrictions apply.

equation to transform the maximum optimization problem into a convex minimum optimization problem

$$\widehat{x}^c_{\mathrm{MAP}} = \operatorname{argmin}\left\{ \begin{array}{l} \sum (y^c - Dx^c)^2 + \dfrac{\sigma^2}{\sigma'^2} \sum \left(x^c_{\mathrm{fine}} - x^c\right)^2 \\ + \dfrac{2\sigma^2}{\beta} U(x^c) \end{array} \right\}. \tag{12}$$

The complicated coefficient can be simply represented by one parameter by assuming $\lambda_1 = (\sigma^2/\sigma'^2)$ and $\lambda_2 = (2\sigma^2/\beta)$, and then (12) can be simply represented as

$$\widehat{x}^c_{\mathrm{MAP}} = \operatorname{argmin}\left\{ \|Dx^c - y^c\|^2 + \lambda_1 \left\|x^c_{\mathrm{fine}} - x^c\right\|^2 + \lambda_2 U(x^c) \right\}. \tag{13}$$

The first term is the spectral constraint term, which indicates that the land-cover classes of solution x^c should be subject to the numerical information provided by the abundance images y^c. The second term is the temporal constraint term, which provides the spatial pattern constraint in the desired scale specific to the research region. The third term is the spatial regularization term, which simulates the distribution pattern of the HRTI x^c and preserves the smoothness of the boundaries between each class.

The spatial regularization term $U(x^c)$ has long been an effective strategy in the MAP model, serving as prior constrained knowledge for probability estimation. Many spatial regularization models have been designed and developed, according to the generalization of the inherent features within the images, i.e., observations with the same property are inclined to be spatially close to each other (the property can be the gray value in natural images or the class label in classification images), the property of adjacent observations changes only gradually and the boundaries between observations tend to be contiguous. Alternative prior models have also been designed for maintaining a regular detail and structure, instead of having a smoothing effect [44]–[47]. In the proposed SST_MAP model, the commonly used TV model [44] is used as the spatial regularization model, since the TV model gives consideration to both boundary smoothness and detail patterns, and is formulated as

$$U(x^c) = \sum_i \sum_j \sqrt{\left|\nabla x^c_h\right|^2 + \left|\nabla x^c_v\right|^2} \tag{14}$$

where ∇x^c_h and ∇x^c_v represent the gradient in the horizontal and vertical directions of image x^c, respectively. In a discrete digital image, the gradient can be computed as

$$\begin{aligned} \nabla x^c_h &= x^c[i+1, j] - x^c[i, j] \\ \nabla x^c_v &= x^c[i, j+1] - x^c[i, j] \\ 0 &\le x^c \le 1 \end{aligned} \tag{15}$$

where λ_1 and λ_2 are the weight coefficients or regularization parameters used to control the balance between the spectral term, the temporal term, and the spatial term. In this article, the two regularization parameters were manually tuned and selected. The results of this analysis are provided in the sensitivity analysis part of Section V. Stochastic gradient descent is adopted to iteratively solve the SST_MAP model presented in (13). The established MAP model should be first transformed into an energy function.

The energy function is then iteratively decreased in the descent direction by updating x^c, where the descent direction can be obtained by the derivation of the energy function

$$\nabla E = 2D^T(Dx^c - y^c) + 2\left(x^c - x^c_{\mathrm{fine}}\right) + \nabla U(x^c) \tag{16}$$

where the derivation of the spatial prior term can be represented as

$$\nabla U(x^c) = \sum_i \sum_j \nabla \cdot \left(\frac{\left[\nabla x^c_h \ \nabla x^c_v\right]^T}{\sqrt{\left|\nabla x^c_h\right|^2 + \left|\nabla x^c_v\right|^2 + \beta}} \right). \tag{17}$$

The x^c at epoch $n+1$ can be represented by the x^c at epoch n, plus a step toward the descent direction, which can be denoted as

$$\begin{aligned} &\hat{x}^c_{n+1} \\ &= \hat{x}^c_n + lr \cdot \left(2D^T(Dx^c - y^c) + 2\lambda_1\left(x^c - x^c_{\mathrm{fine}}\right) + \lambda_2 \nabla U(x^c)\right) \end{aligned} \tag{18}$$

where lr is the learning rate, and the stopping condition is simply set to a fixed iteration number.

The whole architecture of the proposed algorithm is summarized in Fig. 4. The input abundance images y^c and the TCFI x^c_{fine} are applied to build the downsampling observation model and the transition observation model, and are then combined with the spatial regularization model to establish the proposed SST_MAP estimation model. The solution HRTI x^c at time T_2 can then be compared to the previous TCFI x^c_{fine} at time T_1 to derive the LCCD map, which is able to achieve a high spatial resolution by the sub-pixel reconstruction of y^c and a high-frequency temporal resolution by the short time interval between the selected multiscale bitemporal images.

IV. Experiments and Analysis

In this article, four experiments were conducted to validate the proposed SST_MAP model: three synthetic-image experiments and one real-image experiment. In the synthetic-image experiments, the input abundance images were generated by downsampling the reference classification image, and SPM was adopted to restore the synthetic abundance images to the fine-scale classification image. The downsampling process is generally implemented by an averaging filter and the original reference classification image can be used for accuracy assessment of the SPM result. In contrast, in the real-image experiment, the input abundance images were derived from spectral unmixing of the real coarse image, and SPM was adopted to restore the abundance images to the fine-scale classification image. A real fine classification map of the research area was thus required for the accuracy validation of the SPM result. Both the synthetic and real manners are feasible: the synthetic abundance image manner is able to avoid the uncertainty derived from the preprocessing, such as the coregistration (between the coarse and fine images) and spectral unmixing processes, which allows the SPM results to solely reflect the performance of different algorithms. In contrast, the real abundance image manner accounts for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:58:24 UTC from IEEE Xplore. Restrictions apply.

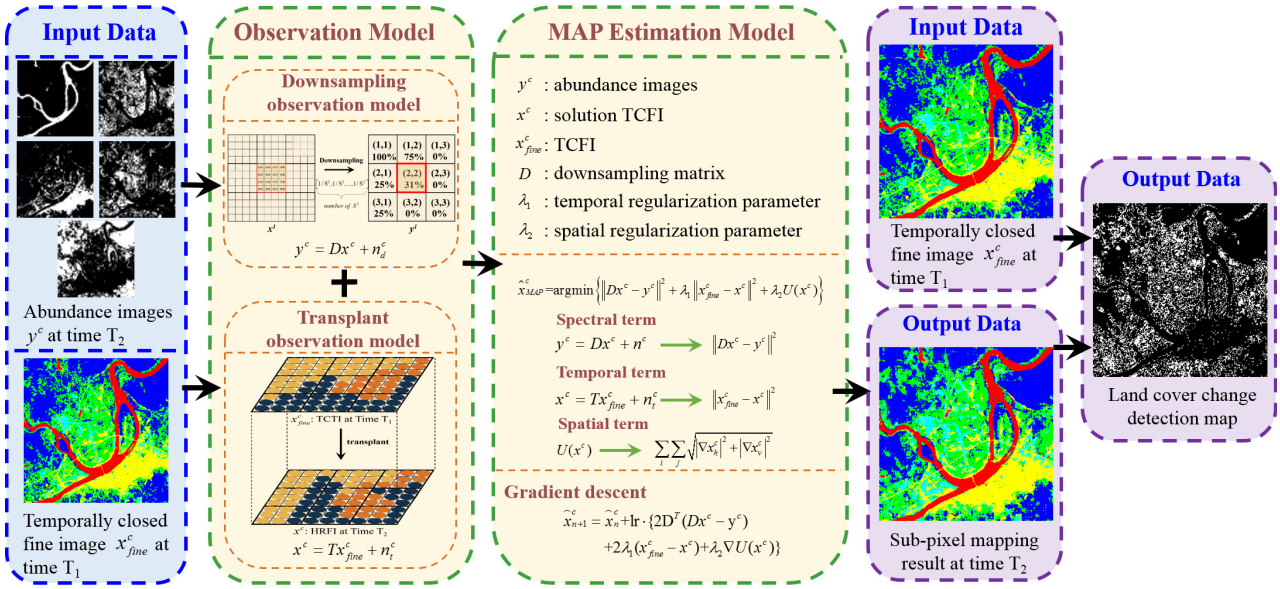


Fig. 4. Flowchart of the SST_MAP model.

the practical demand of directly handling a real coarse image, which can evaluate the adaptability and tolerability to the uncertainty in a real situation.

Traditional SPM methods were conducted for comparison: PSSM [12], CSSM [17], adaptive MAP-based class determination and sub-pixel mapping (AMCDSM) [28], as well as the state-of-the-art spatio-temporal architecture of sub-pixel LCC mapping (SLCCM) [33], super-resolution mapping based on spatio-temporal dependence (SRM_STD) [34], spatio-temporal sub-pixel mapping-based clonal selection (SSMCS) [48], SPM algorithm based on subpixels and pixels spatial dependence, and temporal dependence (SPT) [36], and spatio-temporal geographical evolution mapping (STGEM) [49].

SLCCM is based on a pre-defined LCC rule, which acts as a decision-making strategy for the incorporation of the temporal information and the sub-pixel attribute allocation. SSMCS is an expansion of SLCCM, but the difference is that SSMCS optimizes the objective dependence by the clonal selection algorithm. Meanwhile, SRM_STD, SPT, and STGEM are based on the spatio-temporal dependence model for incorporating the temporal information. The difference is that SRM_STD constructs a transfer matrix between the corresponding sub-pixels in the former fine image and the latter reconstructed image, based on LCC rules, to consider the temporal dependence. SPT uses the sub-pixel-wise correlation to construct the temporal dependence, and the time interval between two temporal images is employed as the decision factor for the temporal image. Finally, STGEM adopts both pixel-wise and sub-pixel-wise correlations for constructing the temporal dependence, and both the time interval and the changed area are considered as the decision factors.

The overall accuracy (OA) and the kappa coefficient (kappa) were selected as the evaluation indicators, and the producer's accuracy of each land-cover class was also employed for the accuracy estimation. The F-score was also adopted for further examination of the reconstruction performance of each individual land-cover class. The population size and the mutation rate in CSSM and SSMCS were set to 100 and 0.5, respectively. In STGEM, the mutation rate was adaptively determined by an adaptive strategy during the generation (the clonal rate α was set to 0.02 and ND was set to 10). In AMCDSM, the regularization parameter was adaptively determined by the adaptive strategy proposed in [28].

A. Experiment 1: Landsat 8 Images (Cili)

In this experiment, bitemporal Landsat 8 images (at 30-m spatial resolution) of the city of Cili, Hunan province, China, were utilized to validate the proposed algorithm. The bitemporal images have six bands (excluding the thermal band 6), 240 lines, and 240 columns, and were acquired on July 13, 2013 and August 7, 2013, as shown in Fig. 5(a) and (c), respectively. Support vector machine (SVM) was applied to the bitemporal images to obtain their thematic maps, with the help of ENVI software. The OAs of the SVM classification for the Landsat 8 images from July 13th and August 7th were 90.36% and 93.91%, respectively, which guaranteed the confidence of the ground truth for the reliable validation of each SPM algorithm. The thematic maps were classified into five categories: water, farmland, vegetation, urban, and bare soil, as shown in Fig. 5(b) and (d). It is worth noting that since the bitemporal images are homologous data, the sensor point spread function (PSF) [50] and geometric effects can be considered to be the same, so the bitemporal images can be compared pixel-by-pixel. The geodetic coordinates of the range of the selected area were (29°4′45.66″–29°8′38.60″N) and (111°0′19.21″–111°4′45.27″E), with the north arrow pointing toward the upside. The synthetic manner was adopted for this data set. Specifically, the thematic map for August 7, 2013, was downsampled to generate the input abundance images by an averaging filter of 4 [as shown in Fig. 5(e)], which indicates that the scale factor of the reconstruction was 4. The SPM methods

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:58:24 UTC from IEEE Xplore. Restrictions apply.

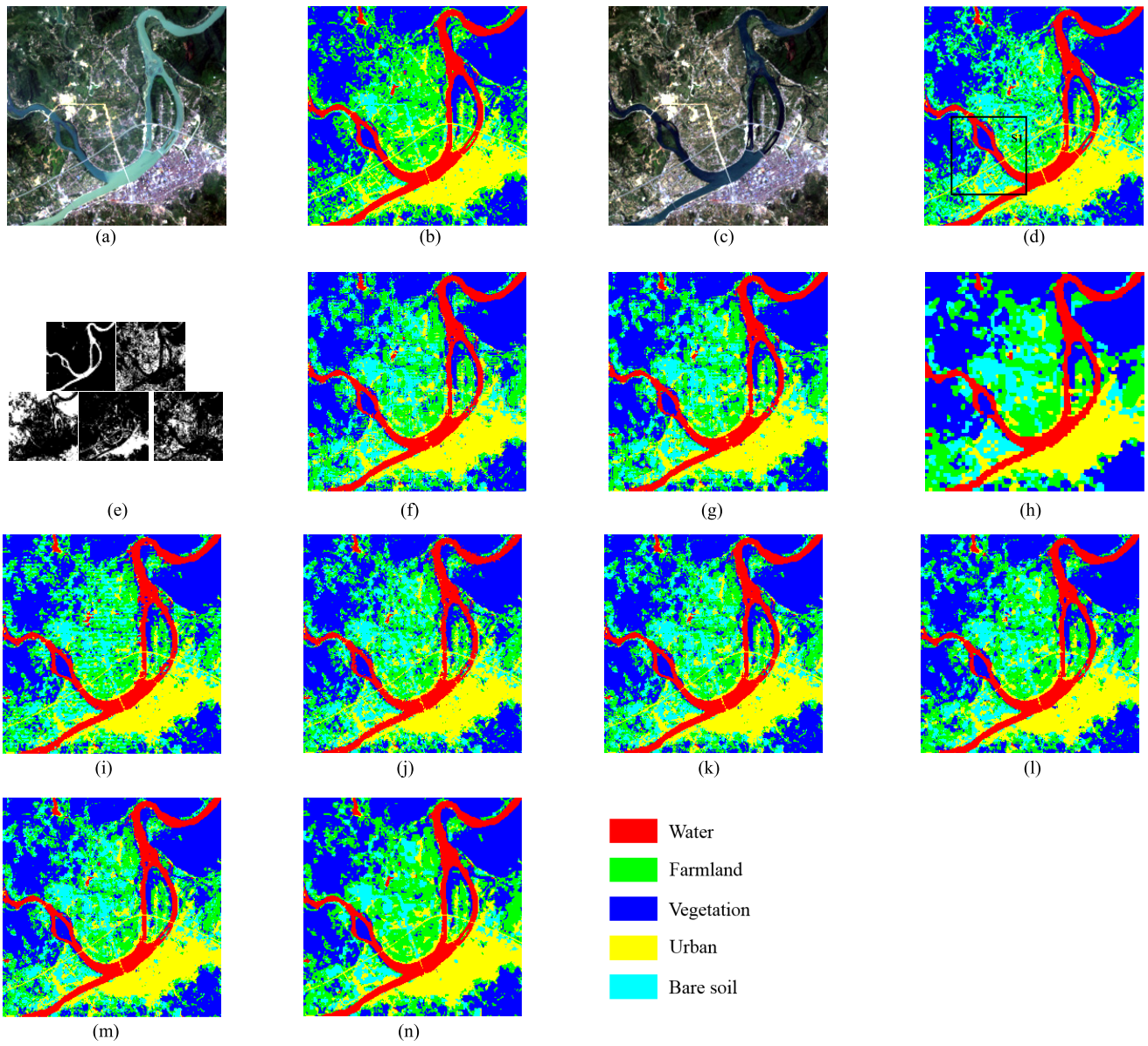


Fig. 5. SPM results for the Landsat 8 image in Experiment 1. (a) Original Landsat 8 image from July 13, 2013 (bands 4, 3, 2) (240 × 240). (b) Original SVM classification result of the July 13, 2013, Landsat 8 image (240 × 240). (c) Original Landsat 8 image from August 7, 2013 (bands 4, 3, 2) (240 × 240). (d) Original SVM classification result of the August 7, 2013, Landsat 8 image (240 × 240). (e) Abundance images of water, farmland, vegetation, urban, and bare soil (60 × 60), respectively. (f) PSSM. (g) CSSM. (h) AMCDSM. (i) SLCCM. (j) SRM_STD. (k) SSMCS. (l) SPT. (m) STGEM. (n) SST_M.

were then implemented to restore the input abundance images to the fine thematic map. It is worth noting that the ratio of the variation was 20.54% between the bitemporal images, which had the purpose of examining the restoration capability of the proposed SST_MAP model for the case of a relatively small variation over a short time interval. Meanwhile, the image size of 240 × 240 with 30-m spatial resolution exposed a dominant homogeneous distribution, which had the purpose of examining the restoration capability of SST_MAP on this occasion.

The SPM results are displayed in Fig. 5(f)–(n). Furthermore, in order to allow a careful inspection of the detailed restoration pattern, zoomed areas are shown in Fig. 6(a)–(k). The quantitative evaluation results of OA, kappa, and producer's accuracy are listed in Table I, and the F-score for each land-cover class is listed in Table II.

From the global and detailed views shown in Figs. 5 and 6, for the single-temporal image-based methods, PSSM performs poorly at restoring linear features and boundaries, such as the missing road over the river (the yellow represents the road) and the zigzagging river bank (the red represents the river). CSSM also performs poorly with linear features, in that the road over the river is also omitted. The performance of AMCDSM is over-smoothed, for the reason that the prior regularization term eliminates the contribution of the original abundance information, making the reconstruction of the detailed structure such as bridges and roads rather disappointing. Meanwhile, the single-image-based SPM algorithms have limitations in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:58:24 UTC from IEEE Xplore. Restrictions apply.

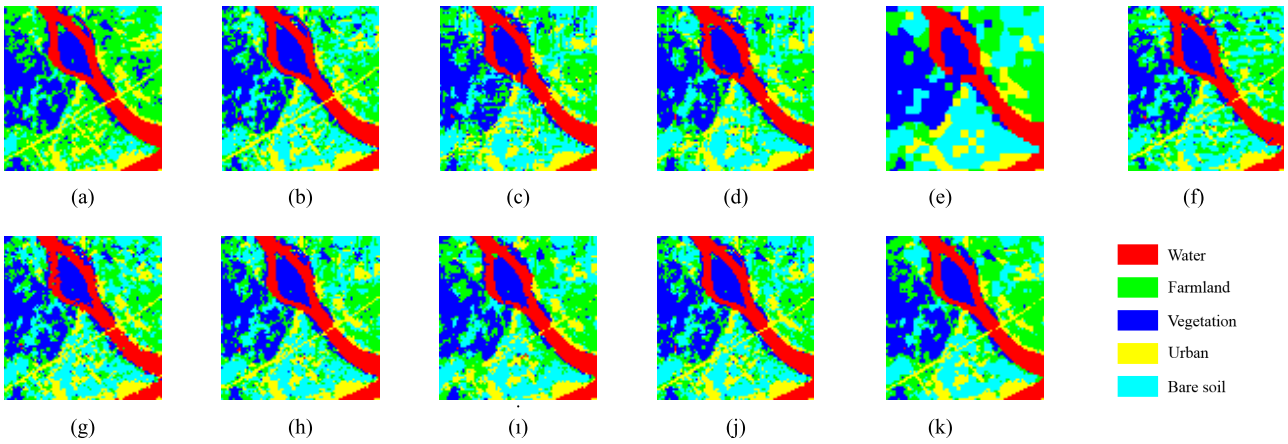


Fig. 6. Zoomed areas of each SPM result. (a) Zoomed-in-view of the TCFI from July 13, 2013. (b) Zoomed-in view of the Landsat 8 reference image from August 7, 2013. (c) Zoomed-in view of the PSSM result. (d) Zoomed-in view of the CSSM result. (e) Zoomed-in view of the AMCDSM result. (f) Zoomed-in view of the SLCCM result. (g) Zoomed-in view of the SRM_STD result. (h) Zoomed-in view of the SSMCS result. (i) Zoomed-in view of the SPT result. (j) Zoomed-in view of the STGEM result. (k) Zoomed-in view of the SST_MAP result.

TABLE I

COMPARISON OF THE SPM RESULTS FOR THE BITEMPORAL LANDSAT 8 IMAGES IN TERMS OF PRODUCER'S ACCURACY, OA, AND KAPPA

Class	Single-temporal image based methods			Multi-temporal image based methods					
	PSSM	CSSM	AMCDSM	SLCCM	SRM_STD	SSMCS	SPT	STGEM	**SST_MAP**
Water	95.60%	95.05%	95.32%	96.86%	94.27%	97.34%	96.12%	97.84%	**98.96%**
Farmland	58.06%	59.26%	62.81%	69.47%	67.18%	68.41%	68.10%	68.68%	**79.10%**
Vegetation	84.07%	85.33%	87.26%	87.64%	89.18%	89.99%	88.17%	90.51%	**91.80%**
Urban	77.13%	78.61%	76.47%	84.17%	86.96%	88.21%	83.34%	90.03%	**89.89%**
Bare soil	55.49%	58.59%	57.33%	63.46%	63.13%	65.75%	57.37%	**65.09%**	53.30%
AA	74.07%	75.37%	75.84%	80.32%	80.14%	81.94%	78.62%	82.43%	**82.61%**
OA	73.61%	75.01%	76.01%	79.98%	80.12%	81.61%	78.68%	82.08%	**83.12%**
Kappa	0.6533	0.6716	0.6839	0.7375	0.7388	0.7584	0.7192	0.7646	**0.7770**

TABLE II

F-SCORE FOR EACH LAND-COVER CLASS

Class	Single-temporal image based methods			Multi-temporal image based methods					
	PSSM	CSSM	AMCDSM	SLCCM	SRM_STD	SSMCS	SPT	STGEM	**SST_MAP**
Water	95.60%	95.05%	90.22%	94.54%	94.27%	97.34%	96.37%	**97.84%**	96.81%
Farmland	58.06%	59.26%	62.53%	66.65%	67.18%	68.41%	65.52%	68.68%	**72.36%**
Vegetation	84.07%	85.33%	85.71%	89.15%	89.18%	89.99%	87.93%	90.51%	**90.84%**
Urban	77.13%	78.61%	78.46%	87.32%	86.96%	88.21%	84.50%	90.03%	**90.38%**
Bare soil	55.49%	58.59%	60.84%	63.70%	63.13%	65.75%	60.22%	65.09%	63.76%

providing enough detailed distribution patterns for a satisfactory restoration.

As for the multitemporal image-based methods, the detailed spatial distributions are better reconstructed, and the noise and uncertainty have been largely suppressed. To be specific, all the multitemporal image-based methods can accommodate the linear features and are able to reconstruct a complete urban road (represented by yellow) over the river, for the reason that they borrow the fine-scale spatial patterns from the available fine image. This is except for SPT, which produces an incomplete and discontinuous urban road, which is due to the inadequate consideration of the spatial correlation between the different-scale pixels for temporal dependence. For the restoration of the discrete features, SLCCM is weak at preserving the buildings (represented by yellow) located within the farmland, while the other methods are able to preserve them. With regard to the overall smoothness, the result of SRM_STD shows slight speckle artifacts, especially on the river, while the results of other methods tend to be smoother, but are missing detailed spatial distribution information. Since only the spatial regularization term is considered in AMCDSM, which has the effect of making the result smooth, and inadequate fine spatial patterns are provided to constrain and balance the smoothing effect, the performance of AMCDSM is over-smoothed and the majority of the intrinsic distribution information is ignored. Therefore, extra fine spatial distribution information from the TCFI is added to balance the spatial regularization term in the proposed SST_MAP model, achieving an appropriate reconstruction of both smooth boundaries and detailed spatial distribution information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:58:24 UTC from IEEE Xplore. Restrictions apply.

The quantitative evaluation results are listed in Table I. The river is relatively easy to distinguish, since it has clear boundaries and distinct spectral features that are separate from the other land-cover classes, and thus the producer's accuracy reaches more than 95%. It is worth noting that August 7th is the beginning of autumn in China, by which time many farmland crops are ripe and being harvested, and thus the bare soil pixels are increased compared to the image from July 13, showing complex, mixed scenes of farmland crops and bare soil. As a result, it is extremely difficult to distinguish the farmland and bare soil, which is consistent with the producer's accuracy results listed in Table I, i.e., around 58% and 57%, respectively. However, the multitemporal image-based methods show a significant improvement for these highly mixed classes, in terms of producer's accuracy. Specifically, for the farmland, they show an improvement of 10.42% compared to the second-best method of STGEM, and the performance is increased by 16.29% over the single-image-based AMCDSM. However, the reconstruction of the bare soil class fails when using SST_MAP in terms of producer's accuracy, which can be explained as follows. The bare soil proportion on July 13 was 6.56% and 16.15% on August 7, within which only 5.55% of the bare soil pixels remained unchanged during this time interval. Thus, the TCFI from July 13 could only provide faithful bare soil distribution information for the unchanged 5.55%, and restoration of the remainder 10.46% out of 16.15% only relied on the spatial regularization model, and the limitation of using only a spatial regularization model resulted in the failure to achieve a good restoration of bare soil.

Notably, the proposed SST_MAP outperforms SLCCM, SRM_STD, and SPT in terms of OA by 3.14%, 3%, and 4.44%, respectively, and outperforms SSMCS and STGEM by 1.51%, and 1.04%, respectively. From the aspect of the F-score for each land-cover class in Table II, the results are consistent with the producer's accuracy. SST_MAP shows a better reconstruction ability for the farmland, vegetation, and urban area classes, showing improvements of 3.68%, 0.33%, and 0.35% over the second-best method of STGEM, respectively. Meanwhile, the results of SST_MAP for the water and bare soil classes are slightly poorer than those of STGEM and SSMCS, by 0.97% and 1.99%, respectively.

Above all, the visual inspection and quantitative inspection confirm the effectiveness of the proposed SST_MAP. Furthermore, this experiment with Landsat 8 images proves that the proposed SST_MAP model is suitable for the reconstruction of large-scale homogeneous landscapes such as vegetation and farmland. The results also confirm the adaptability of SST_MAP to the relatively short-time interval and the small change ratio around 20%, since in this situation, the TCFI can provide more effective and usable spatial patterns for improving the performance of SPM.

B. Experiment 2: QuickBird Images

In Experiment 2, bitemporal QuickBird images covering a 240 × 240 pixel footprint (at a 2.5-m spatial resolution) of the urban area of the city of Shiyan, Hubei province, China, were chosen to demonstrate the performance of the proposed algorithm. The image acquisition times were 2002 [see Fig. 7(a)] and 2004 [see Fig. 7(c)], respectively. As in Experiment 1, the bitemporal QuickBird images were homologous data that could be compared pixel-by-pixel. The coordinates of the QuickBird images were (413616.00–414211.20 m E) and (3712485.60–3713080.80 mN), with the north direction toward the upside. The SVM classification method was again applied to generate the bitemporal land-cover maps [shown in Fig. 7(b) and (d)], which were used for the downsampling and subsequent validation. The OAs of the SVM classification for the QuickBird images from 2002 and 2004 were 96.23% and 97.82%, which guaranteed the confidence of the ground truth for the reliable validation of each SPM algorithm. The land-cover maps were classified into five land-cover classes: impervious surface, vegetation, pond, building, and road. A synthetic strategy was again adopted for Experiment 2. Specifically, the thematic map from 2004 was spatially degraded by a mean filter with a scale factor of 4 to produce the abundance maps shown in Fig. 7(e), and the multiple SPM algorithms were utilized to restore the synthetic abundance maps, as shown in Fig. 7(f)–(n). Zoomed areas are shown in Fig. 8(a)–(k) to allow a careful inspection of the detailed restoration pattern. It is worth noting that the overall ratio of the geographical variation was 43.19%, and thus this case study had the purpose of examining the restoration capability of the proposed SST_MAP model for a relatively long-time interval with considerable changes. Furthermore, the image size of 240 × 240 with a spatial resolution of 2.5-m exposed a much finer scale in each land-cover class than the Landsat 8 imagery. As a result, more intricate and discrete land-cover classes, such as the narrow roads between the buildings, the scattered trees around the buildings, and the ponds in the residential area, could be used to examine the adaptability of the proposed SST_MAP model to detailed, discrete land-cover reconstruction. The quantitative assessment results for the OA, kappa, and producer's accuracy are listed in Table III, and the F-score for each class is listed in Table IV.

According to the zoomed-up visual inspection shown in Fig. 8, distinct differences between SPM algorithms can be observed. The single-temporal image-based methods perform poorly, with obvious speckle artifacts and discontinuous land-cover restoration, among which the result of AMCDSM is again over-smoothed, effacing most of the detailed distribution information. Meanwhile, the multitemporal image-based methods perform much better in preserving the continuity of the linear objects and the consistency between each class. Among the different methods, SPT is weak at preserving linear structures, as in Experiment 1, while SRM_STD, SSMCS, and STGEM perform better in contiguous narrow trail reconstruction. For the proposed SST_MAP model, a continuous distribution can be achieved, such as the continuous road.

However, the distribution within each class is slightly over-smoothed in SST_MAP. This can be attributed to the tradeoff problem between the spatial regularization model and the temporal regularization model. The spatial regularization model tends to remove the artificial speckles as well as the faithful

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:58:24 UTC from IEEE Xplore. Restrictions apply.

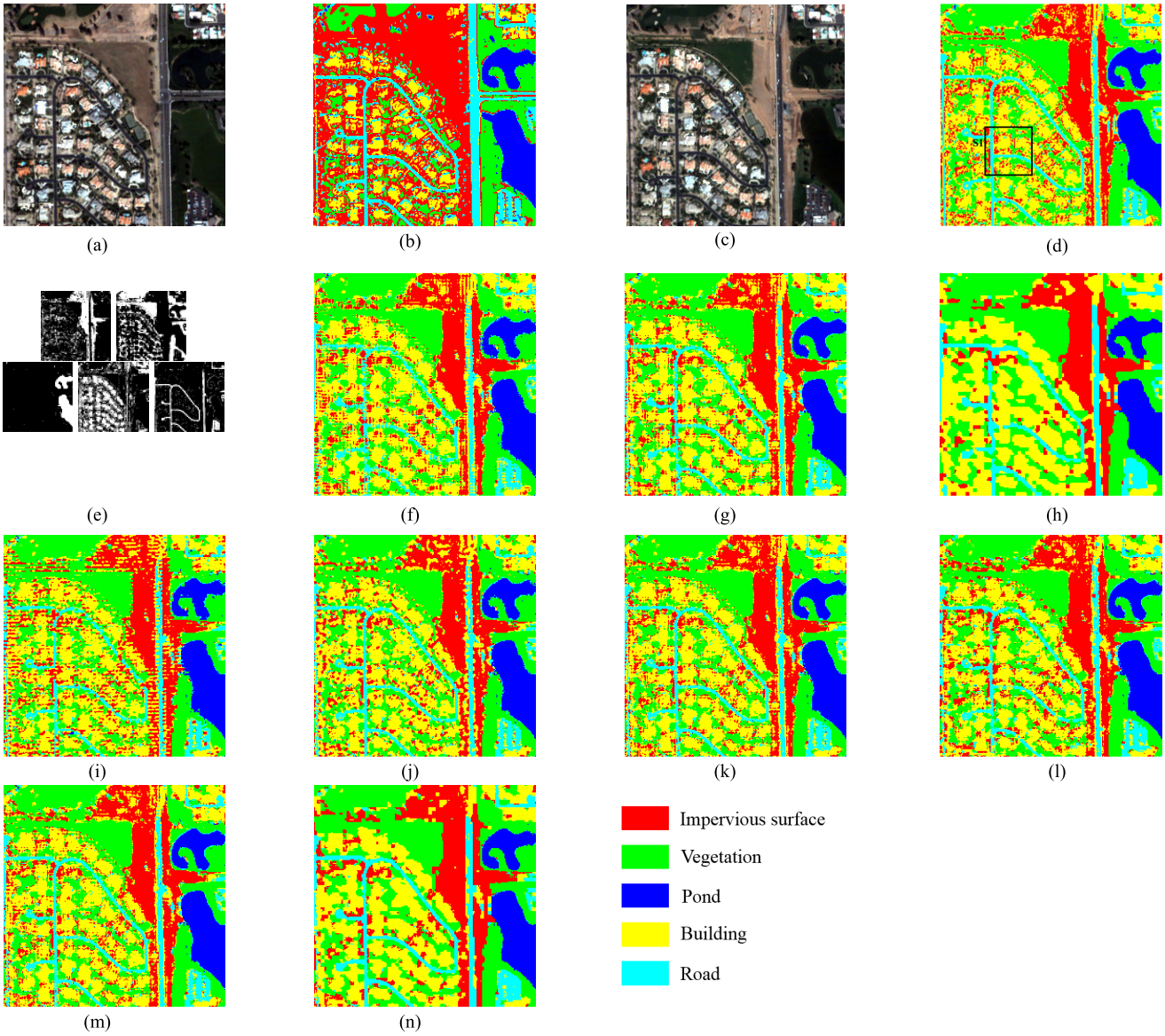


Fig. 7. SPM results for the Hubei QuickBird imagery in Experiment 2. (a) Original QuickBird image from 2002 (bands 3, 2, 1) (240 × 240). (b) Original SVM classification result for the 2002 QuickBird image (240 × 240). (c) Original QuickBird image from 2004 (bands 3, 2, 1) (240 × 240). (d) original SVM classification result for the 2004 QuickBird image (240 × 240). (e) Abundance images of impervious surface, vegetation, pond, building, and road (60 × 60), respectively. (f) PSSM. (g) CSSM. (h) AMCDSM. (i) SLCCM. (j) SRM_STD. (k) SSMCS. (l) SPT. (m) STGEM. (n) SST_MAP.

details and makes the overall visual effect homogeneous. Meanwhile, the temporal regularization model tends to provide the detailed distribution information and may even overwrite the original abundance information with an unfaithful distribution if the change ratio is large during the time interval. Therefore, in this case, the temporal regularization parameter is reduced to avoid this overwriting problem, and thus the final reconstruction map is mostly contributed by the spatial regularization model, and is thus slightly over-smoothed within each class.

For a more detailed verification of the results, a quantitative comparison of the above algorithms is provided in Table III, in terms of the producer's accuracy, OA, and kappa. Since the distribution of the impervious surface, building, and road classes is densely mixed with an intricate spatial structure, this leads to a low producer's accuracy, i.e., around 72%, 70%, and 65%, respectively, for the traditional single-temporal image-based SPM methods. Despite this, the multitemporal image-based SPM methods can better handle the challenge, providing specific fine-scale spatial patterns for detailed information restoration. Among the different methods, the proposed SST_MAP model shows the most improvement, with the accuracy of these three land-cover classes increased by around 5%, 1%, and 13%, respectively. The F-score for each land-cover class in Table IV also shows a similar trend, where SST_MAP shows a better reconstruction ability for the impervious surface class, with an improvement of 1.65% over the second-best method of STGEM, while the results of other land-cover classes are less accurate than the best results by around 1%.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:58:24 UTC from IEEE Xplore. Restrictions apply.

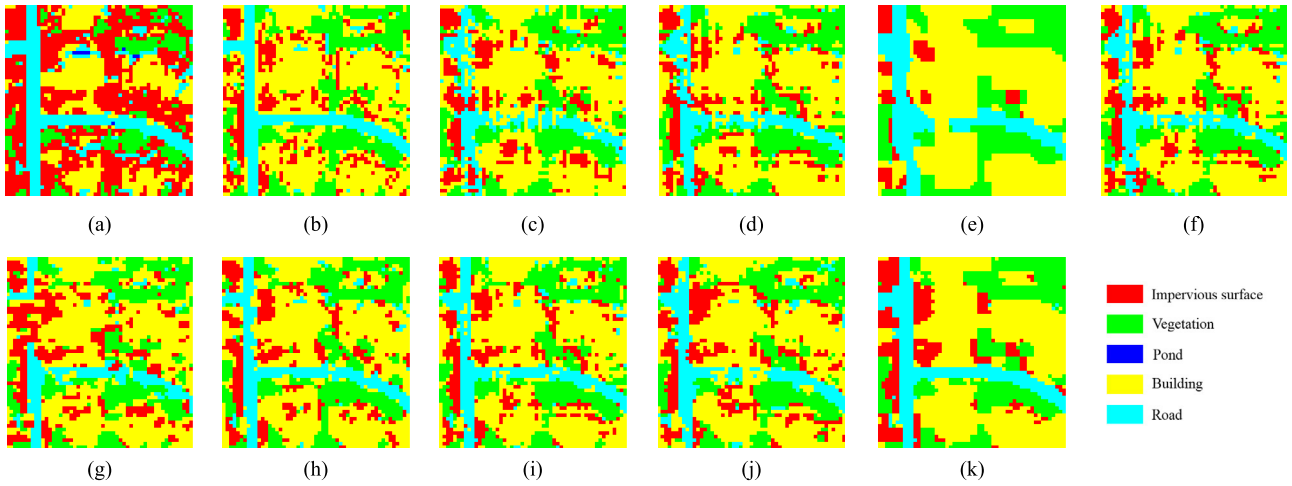


Fig. 8. Zoomed areas of each SPM result. (a) Zoomed-in view of the TCFI from 2002. (b) Zoomed-in view of the reference 2004 QuickBird Image. (c) Zoomed-in view of the PSSM result. (d) Zoomed-in view of the CSSM result. (e) Zoomed-in view of the AMCDSM result. (f) Zoomed-in view of the SLCCM result. (g) Zoomed-in view of the SRM_STD result. (h) Zoomed-in view of the SSMCS result. (i) Zoomed-in view of the SPT result. (j) Zoomed-in view of the STGEM result. (k) Zoomed-in view of the SST_MAP result.

TABLE III

COMPARISON OF THE SPM RESULTS FOR THE BITEMPORAL QUICKBIRD IMAGE IN TERMS OF PRODUCER'S ACCURACY, OA, AND KAPPA

Class	Single-temporal image based methods			Multi-temporal image based methods					
	PSSM	CSSM	AMCDSM	SLCCM	SRM_STD	SSMCS	SPT	STGEM	**SST_MAP**
Impervious surface	71.92%	72.78%	69.67%	70.27%	73.51%	74.17%	73.70%	74.35%	**77.53%**
Vegetation	85.13%	86.54%	88.05%	85.50%	88.84%	88.40%	86.16%	88.33%	**89.38%**
Pond	96.88%	98.15%	97.61%	95.90%	97.86%	98.04%	98.02%	98.20%	**98.38%**
Building	70.58%	72.98%	74.39%	71.30%	74.75%	74.99%	71.84%	**74.91%**	71.68%
Road	65.34%	74.36%	69.34%	68.04%	78.92%	77.89%	76.88%	**79.07%**	78.65%
AA	77.97%	80.96%	79.81%	78.20%	82.78%	82.70%	81.32%	82.97%	**83.12%**
OA	76.78%	79.15%	78.88%	76.99%	80.99%	80.98%	79.97%	**81.10%**	81.01%
Kappa	0.6905	0.7220	0.7174	0.6929	0.7467	0.7465	0.7220	**0.7481**	0.7475

TABLE IV

F-SCORE FOR EACH LAND-COVER CLASS

Class	Single-temporal image based methods			Multi-temporal image based methods					
	PSSM	CSSM	AMCDSM	SLCCM	SRM_STD	SSMCS	SPT	STGEM	**SST_MAP**
Impervious surface	71.92%	72.78%	74.37%	67.90%	73.51%	74.17%	72.26%	74.35%	**76.00%**
Vegetation	85.13%	86.54%	85.35%	85.64%	**88.84%**	88.40%	86.58%	88.33%	87.53%
Pond	96.88%	98.15%	96.46%	96.37%	97.86%	98.04%	98.15%	**98.20%**	97.72%
Building	70.58%	72.98%	73.59%	71.45%	74.75%	**74.99%**	72.78%	74.91%	74.63%
Road	65.34%	74.36%	70.25%	71.97%	78.92%	77.89%	75.63%	**79.07%**	77.51%

As for OA and kappa, the proposed SST_MAP model does not show any superiority over STGEM and SSMCS, obtaining values of 81.01% and 0.7475, respectively. The reason for this is that considerable change occurs during the time interval, and thus fewer fine spatial patterns can be used as a guideline for the current coarse image reconstruction. In spite of this, the computational efficiency of the SST_MAP model is much higher than that of STGEM and SSMCS by almost two orders of magnitude, and it is thus more suitable for practical applications.

C. Experiment 3: Landsat 8 *Images (Shanghai)*

In order to inspect the capability of SST_MAP as well as other participated algorithms for restoring a large-scale area, an extra experiment with bitemporal Landsat 8 images from Shanghai, China, was conducted. The bitemporal images were both resized to 1024 × 1024 pixels, as shown in Fig. 9(a) and (c), and were classified to green vegetation, bare soil, water, and building for the image from December 17, 2018 [see Fig. 9(b)], and an additional cloud class was included for the image from Jan 18, 2019 [see Fig. 9(d)]. The class downsampling operation was implemented to generate the synthesized abundance images with a size of 256 × 256 [i.e., the reconstruction scale factor was 4, as shown in Fig. 9(e)], serving as input for the SPM and SSM algorithms. This research scene is much larger than the original Experiments 1 and 2, and allows us to examine the adaptability of SST_MAP to a large-scale situation. The time interval

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:58:24 UTC from IEEE Xplore. Restrictions apply.

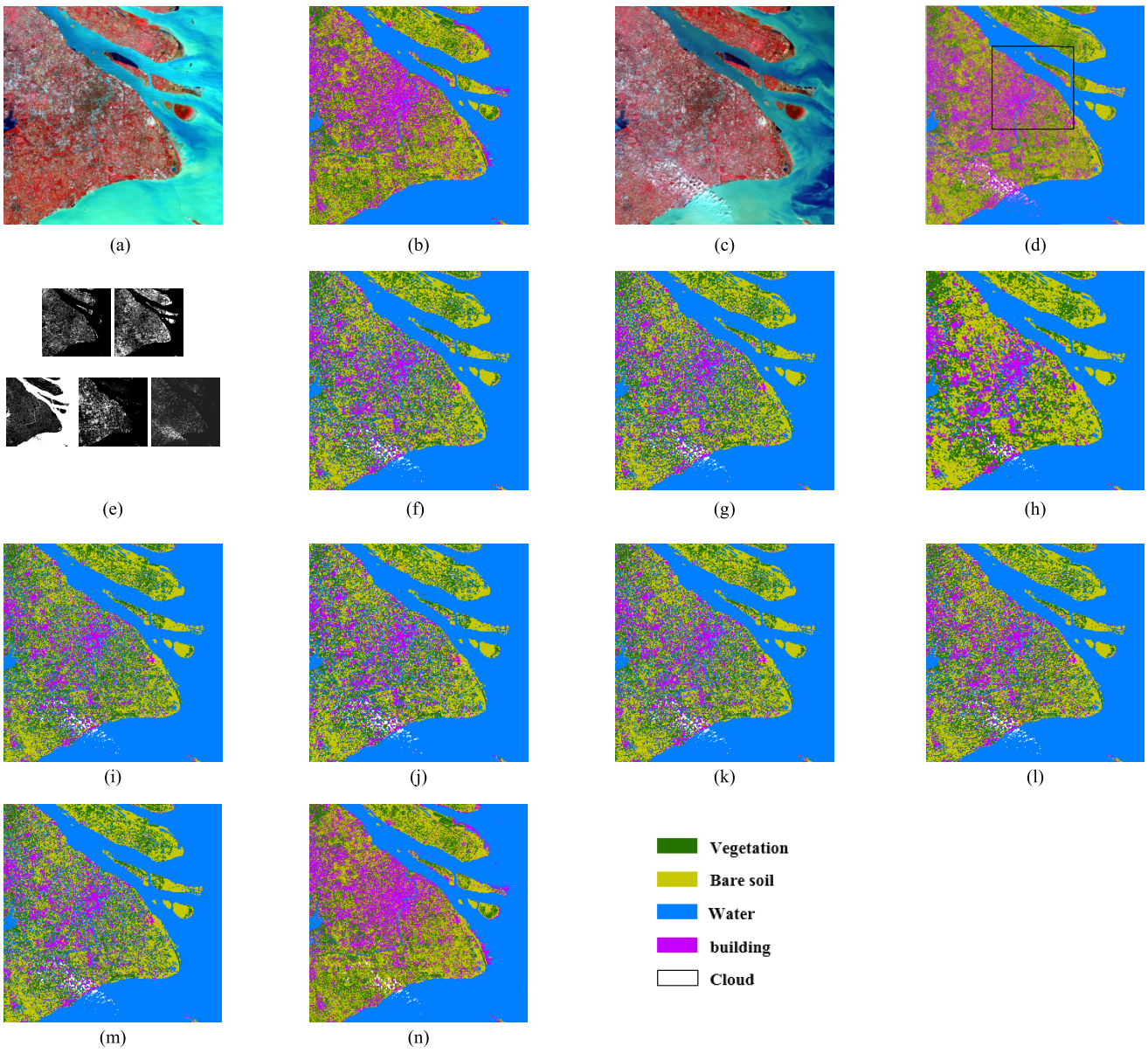


Fig. 9. SPM results for the Shanghai Landsat 8 image in Experiment 3. (a) Original Landsat 8 image from December 17, 2018 (bands 4, 3, 2) (1024 × 1024). (b) SVM classification result for the December 17, 2018 Landsat 8 image (1024 × 1024). (c) Original Landsat 8 image from January 18, 2019 (bands 4, 3, 2) (1024 × 1024). (d) SVM classification result for the January 18, 2019 Landsat 8 image (1024 × 1024). (e) Fraction images of vegetation, bare soil, water, building, and cloud (160 × 160), respectively. (f) PSSM. (g) CSSM. (h) AMCDSM. (i) SLCCM. (j) SRM_STD. (k) SSMCS. (l) SPT. (m) STGEM. (n) SST_MAP.

between the bitemporal images is approximately one month, and the change ratio of this selected research region is 8%.

The SPM and SSM results are shown in Fig. 9(f)–(n), and a zoomed area focusing on the central high-density urban area, as well as the surrounding ocean, was also selected for careful visual inspection, as shown in Fig. 10. The quantitative results are listed in Tables V and VI.

From a visual inspection of Fig. 9 and the zoomed version in Fig. 10, it can be found that the result of the single-temporal-based AMCDSM algorithm is oversmoothed, while the other algorithms produce either fuzzy or deficient heterogeneous details. Among all the algorithms, SST_MAP achieves a better reconstruction. On the whole, the homogeneous areas such as water, bare soil, and vegetation are relatively smooth, while from the zoomed region, it can be seen that the heterogeneous areas, such as buildings with rich structural details, are well distinguished and reconstructed, and are preserved faithfully when compared to the ground truth.

From the quantitative results in Table V, AMCDSM ranks first in the single-image-based SPM methods, obtaining an OA of 71.36%. As for the multitemporal image-based SSM methods, due to the help from the TCFI of December 17, 2018 in providing sub-pixel-scale distribution information, STGEM obtains a slightly increased OA value, while the SLCCM, SSMCS, and SPT methods are almost unchanged in terms of OA. The proposed SST_MAP ranks first in terms

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:58:24 UTC from IEEE Xplore. Restrictions apply.

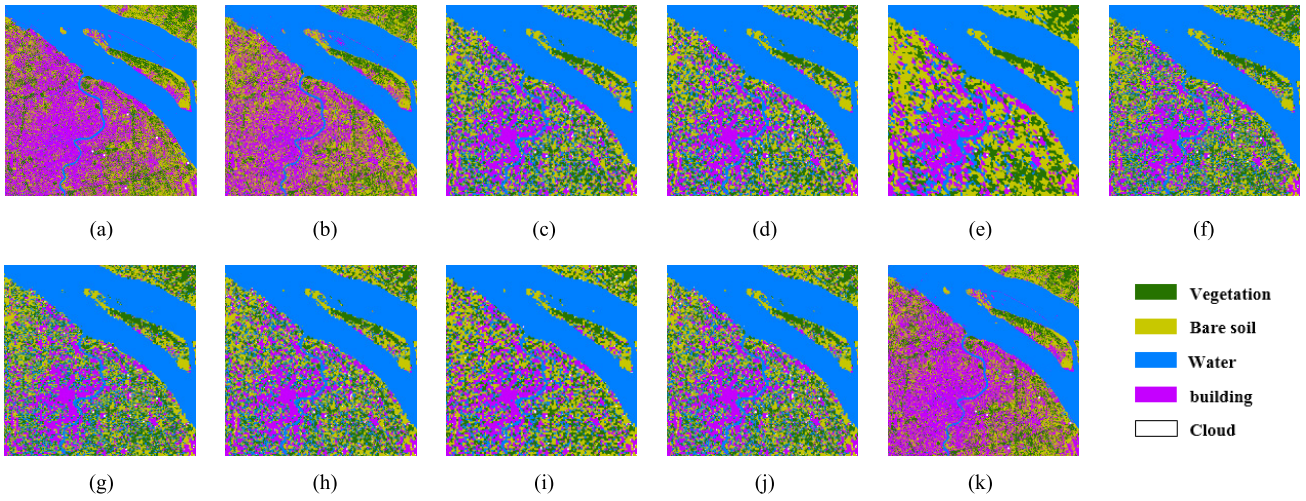


Fig. 10. Zoomed areas of each SPM result. (a) Zoomed-in view of the TCFI from the December 17, 2018, Landsat 8 image. (b) Zoomed-in view of the reference January 18, 2019, Landsat 8 image. (c) Zoomed-in view of the PSSM result. (d) Zoomed-in view of the CSSM result. (e) Zoomed-in view of the AMCDSM result. (f) Zoomed-in view of the SLCCM result. (g) Zoomed-in view of the SRM_STD result. (h) Zoomed-in view of the SSMCS result. (i) Zoomed-in view of the SPT result. (j) Zoomed-in view of the STGEM result. (k) Zoomed-in view of the SST_MAP result.

TABLE V

COMPARISON OF THE SPM RESULTS FOR THE BITEMPORAL LANDSAT 8 IMAGE (SHANGHAI) IN TERMS OF PRODUCER'S ACCURACY, OA, AND KAPPA

Class	**Single-temporal image based methods**			**Multi-temporal image based methods**					
	PSSM	CSSM	AMCDSM	SLCCM	SRM_STD	SSMCS	SPT	STGEM	**SST_MAP**
Vegetation	45.59%	45.61%	51.37%	60.79%	43.57%	55.84%	55.84%	60.96%	**82.78%**
Bare soil	54.39%	54.64%	63.11%	58.56%	52.80%	59.56%	58.55%	62.18%	**73.58%**
Water	97.12%	97.25%	97.62%	97.95%	96.78%	97.97%	97.80%	**98.11%**	97.87%
Building	36.01%	36.08%	38.95%	36.94%	35.16%	39.37%	41.12%	41.77%	**76.05%**
Cloud	39.90%	41.27%	27.50%	37.62%	38.69%	40.42%	39.10%	**39.88%**	27.50%
AA	54.60%	54.97%	55.71%	58.37%	53.40%	58.63%	58.48%	60.58%	**71.56%**
OA	67.58%	67.73%	71.36%	71.31%	66.58%	71.33%	71.28%	73.22%	**84.99%**
Kappa	0.5320	0.5346	0.5882	0.5865	0.5178	0.5863	0.5876	0.6134	**0.7889**

TABLE VI

F-SCORE FOR EACH LAND-COVER CLASS

Class	**Single-temporal image based methods**			**Multi-temporal image based methods**					
	PSSM	CSSM	AMCDSM	SLCCM	SRM_STD	SSMCS	SPT	STGEM	**SST_MAP**
Vegetation	42.96%	43.01%	47.11%	54.36%	40.93	52.58%	51.27%	57.33%	**76.49%**
Bare soil	57.97%	58.25%	62.19%	63.13%	56.70	63.44%	62.09%	66.16%	**75.71%**
Water	88.33%	88.53%	92.62%	89.12%	87.88	89.14%	90.44%	89.29%	**98.41%**
Building	45.54%	45.92%	49.77%	48.42%	44.52	49.93%	51.12%	53.06%	**75.96%**
Cloud	29.03%	26.25%	36.98%	27.87%	25.86	28.65%	28.43%	28.46%	**36.98%**

of OA, greatly surpassing the second-best STGEM method by 11.77%.

From the *F*-score for each class in Table VI, particularly the vegetation class, almost all the SSM algorithms perform better than the single-image-based SPM methods, with STGEM showing an increase from 47.11% (achieved by AMCDSM) to 57.33% (i.e., 10.22%), while SST_MAP shows an increase from 47.11% to 76.49% (i.e., 29.38%), surpassing STGEM by 19.17%. As the research area is Shanghai, it is important to examine the reconstruction performance for the heterogeneous building class. SST_MAP obtains an *F*-score of 75.96%, representing an improvement of 12.9% over the second-best STGEM algorithm, substantiating its reconstruction performance for the heterogeneous class, which is consistent with the visual inspection results.

The low accuracy of the cloud class can be partly attributed to the fact that the TCFI from December 17, 2018 has no cloud class, so that the image cannot provide any cloud distribution information at the sub-pixel scale for the constraint of the reconstruction.

D. Experiment 4: GF-2 and Landsat 8 Images

The three synthetic-image experiments were conducted to purely validate the proposed SST_MAP model; however, in a real situation, the input abundance images are derived from the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:58:24 UTC from IEEE Xplore. Restrictions apply.

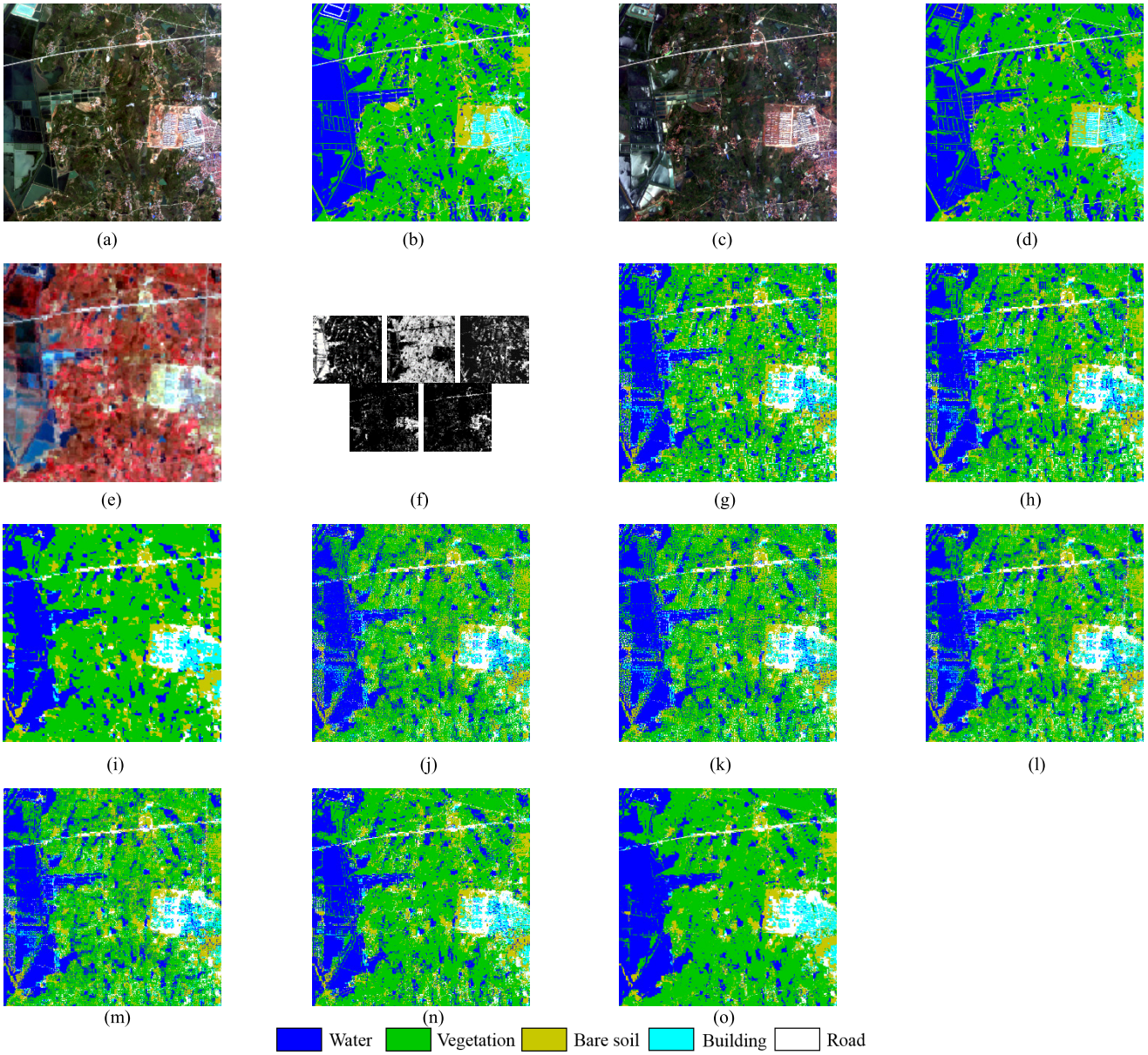


Fig. 11. SPM results for the Wuhan Landsat 8 image in Experiment 3. (a) Original GF-2 image from 2015 (bands 3, 2, 1) (960 × 960). (b) SVM classification result for the 2015 GF-2 image (960 × 960). (c) Original GF-2 image from 2016 (bands 3, 2, 1) (960 × 960). (d) SVM classification result for the 2016 GF-2 image (960 × 960). (e) Original Landsat 8 OLI image from 2016 (bands 3, 2, 1) (160 × 160). (f) Abundance images of water, vegetation, bare soil, building, and road (160 × 160), respectively. (g) PSSM. (h) CSSM. (i) AMCDSM. (j) SLCCM. (k) SRM_STD. (l) SSMCS. (m) SPT. (n) STGEM. (o) SST_MAP.

spectral unmixing of an available coarse image, and significant uncertainty exists in the experimental process, such as the registration error between the coarse and fine images, the spectral unmixing error of the coarse image, and the classification error of the reference fine image. Therefore, a synthetic image does not meet the practical requirements. To further evaluate the effectiveness of the proposed algorithm, an experiment with a real coarse image was also undertaken.

In Experiment 4, bitemporal Gaofen-2 (GF-2) satellite multispectral sensor (MSS) images covering a 960 × 960 pixel footprint (at 4-m spatial resolution) of the urban–rural boundary of Wuhan, Hubei Province, China (30°0′ 14.77″N and 114°2′ 56.46″E), were chosen to inspect the performance of the proposed SST_MAP model in a real situation. The bitemporal GF-2 images were acquired on April 22, 2015 and April 21, 2016 [see Fig. 11(a) and (c), respectively]. Over this period of time, rapid urbanization took place in Wuhan, involving typical Chinese urbanization of the urban–village style. The four spectral bands (blue, green, red, and near-infrared) of the MSS images were selected to participate in the subsequent processing. Meanwhile, a Landsat 8 Operational Land Imager (OLI) multispectral image (at a resolution of 30 m) acquired on March 2, 2016, covering the same field of view as the bitemporal GF-2 images, was chosen as the input real coarse spatial resolution image [shown in Fig. 11(e)]. The Landsat 8 image was downloaded from the USGS, and six bands from band 2 (blue band) to band 7 (short-wave infrared) were selected for the experiment. In the same procedure, the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:58:24 UTC from IEEE Xplore. Restrictions apply.

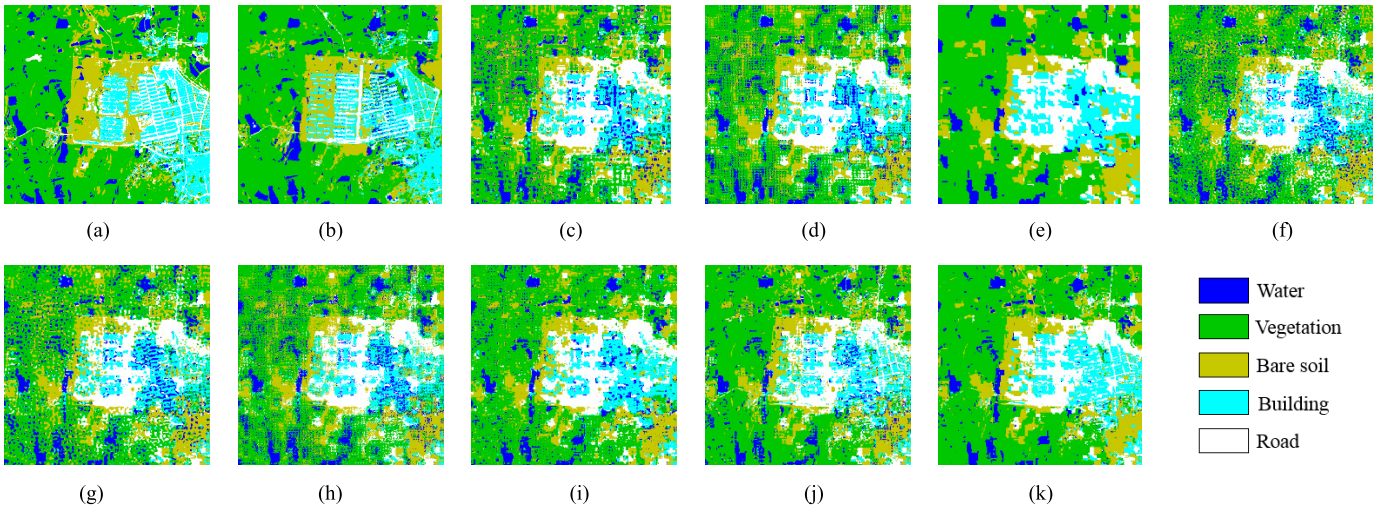


Fig. 12. Zoomed areas of each SPM result. (a) Zoomed-in view of the TCFI from 2015. (b) Zoomed-in view of the reference 2016 GF-2 image. (c) Zoomed-in view of the PSSM result. (d) Zoomed-in view of the CSSM result. (e) Zoomed-in view of the AMCDSM result. (f) Zoomed-in view of the SLCCM result. (g) Zoomed-in view of the SRM_STD result. (h) Zoomed-in view of the SSMCS result. (i) Zoomed-in view of the SPT result. (j) Zoomed-in view of the STGEM result. (k) Zoomed-in view of the SST_MAP result.

bitemporal GF-2 images were classified to land-cover thematic maps with the SVM classification algorithm in ENVI software, which were then used as the available fine image for the multitemporal image-based methods and the reference image for the validation. The OAs of the SVM classification for the GF-2 images from 2015 and 2016 were 97.31% and 96.79%, respectively, which guaranteed the confidence of the ground truth for the reliable validation of each SPM algorithm. Five land-cover classes were found in the land-cover thematic maps, i.e., water, vegetation, bare soil, building, and road [shown in Fig. 11(b) and (d)]. A real-image strategy was adopted in Experiment 4. Firstly, spectral unmixing was applied to the Landsat 8 image from March 2, 2016, to generate the input abundance images. The multiple SPM algorithms were then implemented to restore the abundance images, to estimate the fine-scale thematic image for March 2, 2016, among which the multitemporal image-based SPM algorithms were carried out with the available fine thematic image from April 22, 2015. It is worth noting that the spatial resolution of GF-2 is 4 m, so the GF-2 images were resampled to 5 m to match the Landsat 8 image (30 m) data by six times, i.e., the scale factor S between GF-2 and Landsat 8 was 6. Finally, the thematic image acquired on April 21, 2016, was used as the reference image for the qualitative and quantitative validation of each SPM algorithm. We intended to use the Landsat 8 image from April 21, 2016, as the input coarse image to match the date of the latter GF-2 image; however, heavy rainfall occurred during April to June in 2016, making the Landsat 8 images acquired during this period unsuitable for use. The available Landsat 8 image at the closest time point was the image from March 2, 2016. This was considered acceptable, as the research objective (urbanization) was unlikely to have varied to a large degree over the course of one month. The overall ratio of variation between the bitemporal GF-2 images was 32.56%, and thus this case study had the purpose of examining the restoration capability of the SST_MAP model for a relatively long time interval with moderate variation under real-image experimental conditions, as well as expanding the SPM result to a finer-scale LCCD map for Chinese urbanization characteristics extraction and analysis. Meanwhile, this scene size of 960 × 960 with a spatial resolution of 4-m exposed both a much finer scale for the heterogeneous distribution of buildings and road, as well as the widespread homogeneously distributed classes such as vegetation and water, and could thus be used to examine the adaptability of the proposed SST_MAP model in both aspects.

Coregistration between the GF-2 images and the Landsat 8 image was needed to allow a pixel/sub-pixel comparison, as illustrated in the methodology part above, i.e., each Landsat 8 pixel corresponded to the footprint of 6 × 6 GF-2 sub-pixels, with a scale factor of 6. The coregistration operation was implemented using the ground control point function provided in ENVI software. Several ground control points were selected for the coregistration and the root-mean-square (rms) error was 0.49 sub-pixels, indicating that the registration error was less than a sub-pixel size, thus satisfying the requirement of the subsequent experiment.

The SPM results derived from the multiple algorithms are displayed in Fig. 11(g)–(o). Zoomed areas are shown in Fig. 12(a)–(k), to allow a careful inspection of the detailed restoration pattern. The quantitative assessment results for the OA, kappa, and producer's accuracy are listed in Table VII, and the F-score for each class is listed in Table VIII.

From the results shown in Fig. 11, as well as the zoomed results shown in Fig. 12, it can be seen that the land-cover thematic maps for April 21, 2016, at the GF-2 scale, have been appropriately restored, indicating the effectiveness of the SPM, to a certain extent. For the single-temporal image-based methods, the results of PSSM and CSSM are misrepresented, with disordered artificial speckle within the homogenous vegetation area, unnatural jagged boundaries between classes, and a blurry structure in the heterogeneous urban area, which is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:58:24 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
COMPARISON OF THE SPM RESULTS FOR THE BITEMPORAL PAIR OF GF-2 AND LANDSAT 8 IMAGES IN TERMS OF PRODUCER'S ACCURACY, OA, AND KAPPA

Class	Single-temporal image based methods			Multi-temporal image based methods					
	PSSM	CSSM	AMCDSM	SLCCM	SRM_STD	SSMCS	SPT	STGEM	**SST_MAP**
Water	52.96%	53.06%	56.15%	52.15%	53.15%	53.31%	57.37%	56.77	**63.58%**
Vegetation	60.65%	60.77%	74.64%	60.86%	60.85%	60.91%	70.90%	70.40	**76.23%**
Bare soil	28.52%	28.64%	31.59%	28.38%	28.84%	28.75%	28.54%	29.76	**33.41%**
Building	27.11%	27.03%	30.22%	26.67%	26.68%	26.69%	**31.34%**	29.45	29.55%
Road	31.26%	31.47%	33.71%	31.04%	31.32%	31.37%	31.60%	32.74	**33.53%**
AA	40.10%	40.20%	45.26%	39.82%	40.17%	40.21%	43.95%	43.83	**47.26%**
OA	53.25%	53.34%	62.55%	53.12%	53.42%	53.50%	60.45%	60.08%	**65.35%**
Kappa	0.2736	0.2758	0.3762	0.2717	0.2766	0.2776	0.3538	0.3539	**0.4212**

TABLE VIII
F-SCORE FOR EACH LAND-COVER CLASS

Class	Single-temporal image based methods			Multi-temporal image based methods					
	PSSM	CSSM	AMCDSM	SLCCM	SRM_STD	SSMCS	SPT	STGEM	**SST_MAP**
Water	24.25%	24.18%	35.53%	23.51%	23.38%	23.67%	31.28%	29.71%	**37.44%**
Vegetation	66.29%	66.43%	74.86%	66.46%	66.51%	66.58%	72.81%	73.05%	**76.58%**
Bare soil	53.13%	53.29%	59.27%	52.63%	53.38%	53.51%	59.02%	58.49%	**64.93%**
Building	23.30%	23.45%	28.24%	23.06%	23.61%	23.52%	26.65%	26.61%	**29.23%**
Road	17.76%	17.68%	23.30%	17.72%	17.93%	17.85%	20.95%	20.76%	**24.20%**

mainly due to the uncertainty propagated from the spectral unmixing process, which subsequently misleads the SPM process. In contrast, the results of AMCDSM are much more satisfactory, with clear and smooth boundaries between classes and no salt-and-pepper noise. This is due to the inclusion of the spatial prior term for regularizing the ill-posed SPM problem, which has the effect of balancing the uncertainty of the spectral unmixing. However, detailed structural information on the buildings is missing.

With respect to the multitemporal image-based methods, SLCCM provides an unsatisfactory restoration result compared to its performance in the synthetic-image experiments, for the reason that the disturbance from the spectral unmixing error limits the effectiveness of the available fine image, resulting in the LCC rules failing to extract useful fine spatial distribution information on the restoration. This is also the reason for the failure of SRM_STD and SSMCS, the results of which are full of salt-and-pepper noise. As for SPT and STGEM, the noise is slightly reduced, compared to SLCCM, SRM_STD, and SSMCS; however, the boundaries between classes are blurry and the detailed structure of the urban area is disordered. In the results of SST_MAP, the fuzzy artificial speckle is largely eliminated when compared to the aforementioned SPM algorithms, and the results contain clear and contiguous land-cover boundaries. The homogenous vegetation area is smooth, and the relatively detailed structures of the buildings and roads are reconstructed well in the heterogeneous urban area.

From the quantitative inspection results listed in Table VII, it can be seen that all the accuracies are generally lower than the synthetic manner in the above experiment. One reason for this is the uncertainty and ambiguity introduced from spectral unmixing, where the land-cover thematic information in the generated abundance images is insufficient for interpretation. From the abundance images, it can be seen that the detailed structures of the building and the road are almost lost, and the classes are blended together with blurry boundaries. Another reason is the extremely high resolution of GF-2, which allows an intricate structure of detailed land cover to be displayed, making it difficult to reconstruct.

From the producer's accuracy results listed in Table VII, the water and vegetation land-cover classes form the homogenous areas, which are relatively easy to restore, and thus their producer's accuracy is higher, reaching 57.37% and 70.90% for the second-best SPT, respectively. The proposed SST_MAP model further improves the accuracy for water and vegetation by 6.15% and 5.33%, respectively, reaching 63.58% and 76.23%, thereby preserving the smoothness of the homogenous areas. In contrast, the building and road classes form the heterogeneous areas, demonstrating an intricate distribution, and thus the producer's accuracy is relatively disappointing. Despite this, the proposed SST_MAP shows gains of 0.79% for the road classes, compared to the second-best STGEM algorithm, indicating the advantage of SST_MAP in preserving a detailed structure. With respect to the OA and kappa, most of the single-temporal image-based SPM methods only reach around 53% and 0.27, respectively, except for AMCDSM, which achieves 62.55% and 0.3762. The reason is that spatial regularization terms of AMCDSM provides a smooth solution for land covers, which has a good effect for homogeneous area restoration like vegetation and water and contributes a lot in the OA and Kappa improvement.

As for the multitemporal image-based methods, the assessment of SLCCM, SRM_STD, and SSMCS is compromised, reaching only around 53%, but STGEM and SPT

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:58:24 UTC from IEEE Xplore. Restrictions apply.

show significant improvement, reaching 60.08%/0.3539 and 60.45%/0.3538, respectively. In spite of this, the proposed SST_MAP further improves on the OA of the second-best SPT methods by 4.9%, reaching 65.35%. This is down to the regularization capability and the expandability of the MAP framework. The regularization capability is able to regularize and constrain the ill-posed SPM problem, by the effective use of the spatial regularization term. The expandability allows the addition of the temporal term, to consider the spatial distribution of the multitemporal fine image. In this way, the three dimensions are mutually balanced and mutually constrained, forming the joint SST_MAP model. The proposed SST_MAP achieves a satisfactory effect for both smoothness in the homogeneous areas and a detailed structure in the heterogeneous areas, which seems contradictory for the traditional SPM and the state-of-the-art spatio-temporal SPM algorithms. Furthermore, the experimental results confirm the effective use of the TCFI with a one-year interval and a 32% change ratio from the coarse Landsat 8 image by the MAP model, which can realize the full potential of the fine temporal information. Nevertheless, the proposed SST_MAP model has limitations in restoring the fine-scale, discretely distributed buildings from the fuzzy abundance images, as shown in the result map of SST_MAP within the zoomed inspection region; the structural distribution information on the building was lost, and thus may not meet the practical requirement. Furthermore, SST_MAP has limitations in restoring the changed areas; the newly constructed buildings in 2016 have not been very well extracted, from a visual inspection. This is because the TCFI cannot provide any instructional distribution information on the changed areas.

In order to look into the characteristics of the urbanization in Wuhan between 2015 and 2016, the restored thematic map was compared with the fine image to yield the LCCD map. In this way, SST_MAP improves the scale of the coarse image to match the available TCFI, solving the tradeoff problem of LCCD and achieving both a fine spatial scale and high-frequency temporal LCCD observation. The reference LCCD map and the estimated LCCD map are shown in Fig. 13. The reference LCCD map was obtained by simply comparing the GF-2 image from 2016 with the GF-2 image from 2015, and the estimated LCCD map was generated by simply comparing the SST_MAP result with the GF-2 image from 2015. As can be seen in the LCCD map, the main variation trajectory between the two temporal images is focused on the following regions. The trend from bare soil to building is represented by the sienna color, and the light red color in the building zone represents the trend from bare soil to road, clearly illustrating the urban expansion process. The trend from vegetation to bare soil is represented by the maroon color, indicating the reclamation of vegetation. However, the trend from water to vegetation, represented by the coral pink color, is lost in the estimated LCCD map, due to the incomplete reconstruction of the vegetation areas in 2016 by SST_MAP. The trend from building to road, represented by the yellow color, and the trend from building to bare soil represented by the blue color may also be mistaken. Despite this, from the reference LCCD map, the trend from water to vegetation indicates the widespread growth of phytoplankton in aquaculture ponds neglected for a long period of time. As this study area was taken at the urban–rural boundary, these main trajectories all imply the Chinese-style urbanization of the urban–village style, i.e., population movement from the agriculture and aquaculture industries to the urban-related industries. These variation trajectories are appropriately reconstructed by the proposed SST_MAP, to a certain degree, suggesting that SST_MAP has the capability of extracting useful temporal information from multitemporal images, and it provides an alternative, promising approach for LCCD.

E. Computational Complexity

In order to further examine the performance in terms of practical effectiveness, the computational complexity, as well as the computation time, is considered in this section. Although these two metrics are similar, they focus on different aspects. On the one hand, the computational complexity is more convincing than the computation time since it can purely represent the essence of the algorithm, regardless of uncertainties, such as the effectiveness of the coding, the programming platform, the configuration of the user's PC, etc. On the other hand, the computation time can give a more comprehensive and intuitive quantitative assessment of the computational complexity of the algorithm, taking account of all the actual software and hardware conditions in a practical situation.

The computational complexity of SST_MAP consists of two parts: 1) initialization of the fine-scale thematic map x^c using interpolation and 2) stochastic gradient descent for the optimization of x^c. The number of categories is C, the size of the abundance image y^c is $M \times N$, the reconstruction scale factor is S, and the number of gradient descent iterations is T. The interpolation process simply executes one coarse pixel by one coarse pixel, and thus the computational complexity can be represented as $O(M \times N)$. For the process of gradient descent, the spectral term, the spatial regularization model, and the temporal regularization model are calculated simultaneously for each class in each epoch, and the total computational complexity can be represented as $O(M \times N \times C \times T)$. As the interpolation and gradient descent procedure are independent, the total computational complexity can be represented as the summation of these two parts, i.e., $O(M \times N + M \times N \times C \times T)$.

We also recorded the computation times of different algorithms for the synthetic-image experiment of Experiment 1. It is worth noting that we only compared the algorithms involving an optimization process, i.e., CSSM, SSMCS, SRM_STD, SPT, and STGEM, which are comparable (SSMCS and STGEM use the clonal selection algorithm for optimization and SRM_STD and SPT use simulated annealing for optimization). The running times of different algorithms were recorded using a PC with an Intel Core i7-4550U CPU @ 1.50-GHz and 8.00-GB RAM, using MATLAB R2013a. All the parameters and iteration times in the corresponding algorithms were set appropriately to obtain the best accuracy (the number of iterations for the evolutionary

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:58:24 UTC from IEEE Xplore. Restrictions apply.

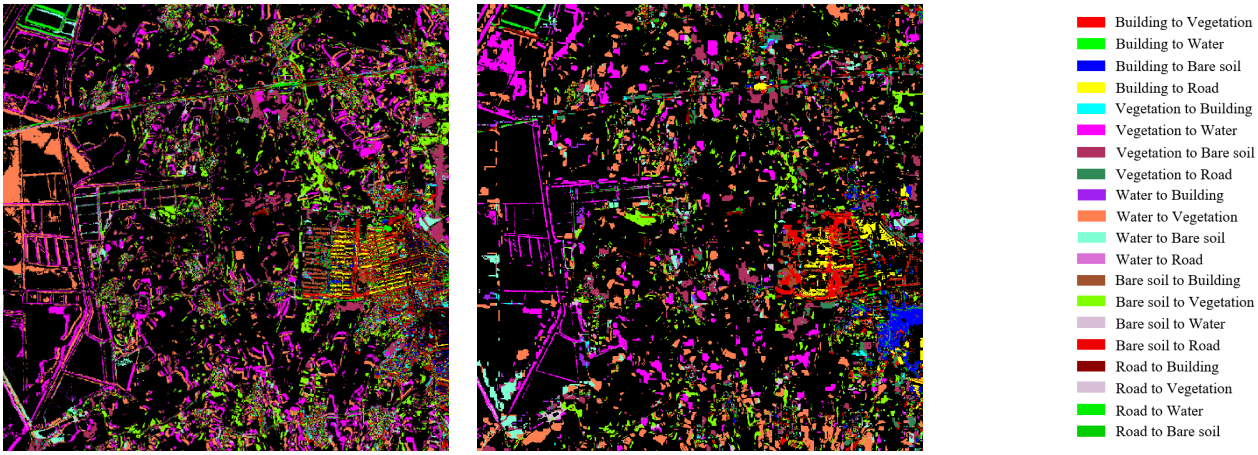


Fig. 13. LCCD maps from 2015 to 2016. (a) LCCD map between the two GF-2 images as the reference. (b) LCCD map between the GF-2 image from 2015 and the reconstructed image for 2016.

algorithm was 100 in CSSM, SSMCS, and STGEM, and the number of iterations for the simulated annealing was 1000 in SRM_STD and SPT). The running times of different algorithms are listed in Table IX. From the results, it can be seen that the proposed SST_MAP model is the most efficient approach, outperforming the other optimization methods by two orders of magnitude. The main reason for this is that the gradient descent optimization is globally optimized, and it is more simple and effective than the evolutionary computation and simulated annealing optimization.

V. Sensitivity Analysis

In the proposed SST_MAP model, λ_1 and λ_2 are the crucial regularization parameters since they determine the proportional contributions of the spectral term, temporal term, and spatial term, and only the proper equilibrium of these three dimensions can lead to the optimal solution. Generally speaking, λ_1 is the temporal regularization parameter determining the contribution of the temporal regularization term and λ_2 is the spatial regularization parameter determining the contribution of the spatial regularization term. If the selection of the spatial regularization parameter is unsuitably small, the speckle artifacts will not be suppressed; however, if the selection of the spatial regularization parameter is unsuitably large, the image will be over-smoothed and detailed structural information will be lost. To achieve both smoothness in the homogeneous areas and structural detail in the heterogeneous areas, the temporal regularization term is adopted to balance the smoothing effect of the spatial term by providing detailed spatial structure and patterns. In the same way, if the selection of the temporal regularization parameter is too small, insufficiently available fine-spatial patterns will be exploited to provide structural details, while if the selection of the temporal regularization parameter is too large, the final SPM result will be overwritten by the TCFI, leading to a false reconstruction.

In order to study the synergistic effect of the two regularization parameters, the bitemporal Landsat 8 images used in Experiment 1 and the bitemporal QuickBird images used in Experiment 2 were selected. The number of iterations in the gradient descent was set to a constant of 41, according to the convergence condition of the MAP optimization process for the experimental data sets. According to the statistical results, the overall ratio of variation between the bitemporal images in Experiment 1 was 20.54%, and this was 43.19% in Experiment 2. Therefore, more fine spatial patterns from the TCFI could be provided in Experiment 1, so the temporal regularization parameter should be larger, while fewer fine spatial patterns could be provided in Experiment 2, so the temporal regularization parameter should be smaller. For this reason, the value of the spatial regularization parameter was set to [0, 2] with a step size of 0.1. The difference was that the value of the temporal regularization parameter was set to [0.01, 0.1] with a step size of 0.01 in Experiment 1, while in Experiment 2, the temporal regularization parameter was set to [0.001, 0.01] with a step size of 0.001. The corresponding OAs of the proposed SST_MAP model with different configurations of λ_1 and λ_2 are displayed in the form of curve graphs in Fig. 14. As shown in Fig. 14, all the curves show an increase to the peak and then fall away. This phenomenon is consistent with the mechanism of the proposed SST_MAP model. With the increase of the spatial regularization parameter in the x-axis, the smoothing effect tends to establish contiguous boundaries, enhances the connections of the same classes, and eliminates speckle artifacts, resulting in an increase in OA. However, if the spatial regularization parameter is in excess of the optimum value, the over-smoothing effect will efface the detailed structure and compromise the accuracy. On the other hand, more detailed spatial patterns can be provided to help with the SPM restoration, and thus the OA is increased.

Therefore, the optimum values of λ_1 and λ_2 in Experiment 1 were 0.2 and 0.01, respectively, while the values were 0.2 and 0.008 in Experiment 2. It is worth noting that when the temporal regularization parameter was set in the range of 0.01–0.1 in Experiment 2, the OA continuously decreased, which confirms our original conclusion, i.e., more fine spatial patterns from the TCFI could be provided in Experiment 1, while fewer fine spatial patterns could be provided in Experiment 2.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:58:24 UTC from IEEE Xplore. Restrictions apply.

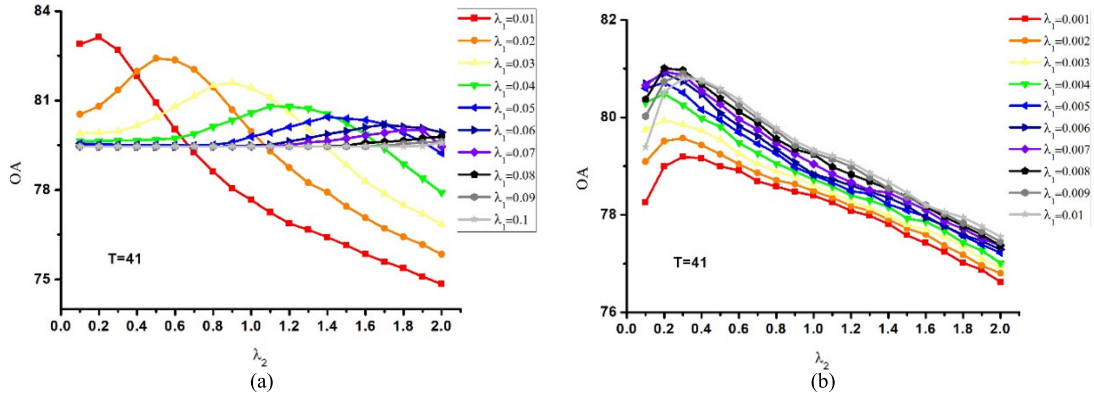


Fig. 14. Sensitivity analysis of SST_MAP in relation to the temporal regularization parameter λ_1 and spatial regularization parameter λ_2. (a) SPM results in relation to the two regularization parameters for the Landsat 8 image in Experiment 1. (b) SPM results in relation to the two regularization parameters for the QuickBird image in Experiment 2.

TABLE IX
RUNNING TIMES OF DIFFERENT ALGORITHMS FOR THE LANDSAT 8 IMAGE IN EXPERIMENT 1

Algorithm	CSSM	AMCDSM	SSMCS	SRM_STD	SPT	STGEM	SST_MAP
Time (s)	2687.03	113.56	2327.67	3035.34	2453.78	1802.00	116.46

VI. CONCLUSION

In this article, the joint SST_MAP estimation model for subpixel mapping has been proposed, to regularize the ill-posed SPM problem and constrain the solution to a more appropriate one. In the MAP estimation model, the SST dimensions are simultaneously considered, i.e., the spectral term constrains the proportion of each land-cover class; the spatial term smooths the boundary between each class, depresses the noise, and balances the uncertainty of the abundance maps; and the temporal term provides an extra specific fine spatial pattern constraint for preserving the detailed spatial structure and neutralizes the over-smoothing effect of the spatial term. In this way, the three dimensions are mutually balanced and mutually constrained, and are able to reach an equilibrium point, achieving a restoration of both smooth areas for homogeneous land-cover classes and a detailed structure for heterogeneous land-cover classes.

Furthermore, the newly developed temporal regularization model can fully exploit the spatio-temporal information, incorporating the specific spatial distribution patterns from the fine image to further regularize the ill-posed SPM problem. This incorporation is based on the expansibility of the MAP model, in that a TCFI can be easily incorporated into the joint processing and simultaneous optimization. In addition, the developed temporal regularization term is able to capture the fine-scale variation between two temporal images in heterogeneous areas.

Despite all this, the SST_MAP model has some limitations. Its result is impacted by the regularization parameters, and a slight deviation from the optimal parameter settings will mislead the mapping results, either to be over-smoothed when the spatial regularization term conquers the temporal term, or to be overwritten when the temporal term conquers the spatial regularization term. For the latter case, when the temporal term dominates the reconstruction, the excessive use of the TCFI will destroy the original abundance information. Sometimes, in order to achieve a high OA and a satisfactory visual result, the original abundance constraint must be partly sacrificed. Furthermore, the proposed SST_MAP model has limitations in restoring the changed areas, since the TCFI cannot provide any instructional distribution information on the changed areas, i.e., if brand new classes occurred in the later temporal, the TCFI could provide nothing for its spatial distribution.

Three synthetic-image experiments were undertaken to validate the proposed SST_MAP model, with bitemporal Landsat 8 images and QuickBird images, and most of the results confirmed the superiority of the proposed algorithm, both qualitatively and quantitatively, except for Experiment 2. The reason is that excessive change occurred during the time interval, and thus fewer fine spatial patterns can be used for the constraint. These experiments specified the suitable range of the change ratio for the proposed SST_MAP, if the change ratio is lower than 30%, the SST_MAP is superior to other SSM algorithms, while if the change ratio is larger than 40%, the SST_MAP may be inferior to other SSM. Besides, the lesser the change ratio (such as Experiment 3), the more superiority the SST_MAP has over other SSM algorithms. A real-image experiment was also designed to further examine the performance of SST_MAP in a practical situation, using a GF-2 and Landsat 8 image pair. The real-image experiment considered the registration error between the coarse and fine images, the classification error of the fine image, the sensor PSF, atmospheric and geometric effects, classification or spectral unmixing errors, etc. Despite the rigorous conditions, the proposed SST_MAP model largely suppressed the artificial speckle noise and uncertainty, extracting effective finer-scale distribution patterns from the available fine image to help the reconstruction and balance the uncertainty and ambiguity, generating a smooth result. Furthermore, in Experiment 4, the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:58:24 UTC from IEEE Xplore. Restrictions apply.

proposed SST_MAP model was adopted to improve the spatio-temporal resolution and generate an LCCD map for LCC information extraction and analysis. The spatial resolution was improved by the SST_MAP model and the temporal resolution was improved by the Landsat image (16-day revisit period), compared to the GF-2 image (69-day revisit period). Detailed LCCD information was appropriately reconstructed by the proposed SST_MAP model, suggesting typical Chinese-style urbanization of the urban–village style in the study area of Wuhan. This case study confirmed that SST_MAP has the capability of extracting useful change information from multitemporal images and can provide an alternative, promising approach for fine-spatial and temporal-scale LCCD.

ACKNOWLEDGMENT

The authors would like to thank the Editor, Associate Editor, and anonymous reviewers for their helpful comments and suggestions that improved this article. They would also like to thank Prof. Q. Wang from Tongji University for providing the code of SPT algorithm and Prof. F. Ling from the Institute of Geodesy and Geophysics, Chinese Academy of Sciences, for providing the code of SRM_STD, which were used in experimental part.

REFERENCES

[1] P. Fisher, "The pixel: A snare and a delusion," *Int. J. Remote Sens.*, vol. 18, no. 3, pp. 679–685, 1997.

[2] R. S. Lunetta, J. F. Knight, J. Ediriwickrema, J. G. Lyon, and L. D. Worthy, "Land-cover change detection using multi-temporal MODIS NDVI data," *Remote Sens. Environ.*, vol. 105, no. 2, pp. 142–154, Nov. 2006.

[3] X. Wang, Y. Zhong, L. Zhang, and Y. Xu, "Spatial group sparsity regularized nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6287–6304, Nov. 2017.

[4] Y. Zhong, X. Wang, L. Zhao, R. Feng, L. Zhang, and Y. Xu, "Blind spectral unmixing based on sparse component analysis for hyperspectral remote sensing imagery," *ISPRS J. Photogram. Remote Sens.*, vol. 119, pp. 49–63, Sep. 2016.

[5] J. Zhao *et al.*, "Spectral-spatial classification of hyperspectral imagery with cooperative game," *ISPRS J. Photogram. Remote Sens.*, vol. 135, pp. 31–42, Jan. 2018.

[6] J. M. Bioucas-Dias *et al.*, "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 354–379, Apr. 2012.

[7] P. M. Atkinson, M. E. J. Cutler, and H. Lewis, "Mapping sub-pixel proportional land cover with AVHRR imagery," *Int. J. Remote Sens.*, vol. 18, no. 4, pp. 917–935, 1997.

[8] G. M. Foody, "Sub-pixel methods in remote sensing," in *Remote Sensing Image Analysis: Including The Spatial Domain*. Amsterdam, The Netherlands: Springer, 2006, pp. 37–49.

[9] P. M. Atkinson, "Issues of uncertainty in super-resolution mapping and their implications for the design of an inter-comparison study," *Int. J. Remote Sens.*, vol. 30, no. 20, pp. 5293–5308, 2009.

[10] L. Zhang, K. Wu, Y. Zhong, and P. Li, "A new sub-pixel mapping algorithm based on a BP neural network with an observation model," *Neurocomputing*, vol. 71, nos. 10–12, pp. 2046–2054, Jun. 2008.

[11] K. C. Mertens, B. De Baets, L. P. C. Verbeke, and R. R. De Wulf, "A sub-pixel mapping algorithm based on sub-pixel/pixel spatial attraction models," *Int. J. Remote Sens.*, vol. 27, nol.15, pp. 3293–3310, Sep. 2006.

[12] P. M. Atkinson, "Sub-pixel target mapping from soft-classified, remotely sensed imagery," *Photogramm. Eng. Remote Sens.*, vol. 71, no. 7, pp. 839–846, 2005.

[13] A. J. Tatem, H. G. Lewis, P. M. Atkinson, and M. S. Nixon, "Super-resolution land cover pattern prediction using a Hopfield neural network," *Remote Sens. Environ.*, vol. 79, pp. 1–14, 2002.

[14] Y. Zhong, A. Ma, Y. S. Ong, Z. Zhu, and L. Zhang, "Computational intelligence in optical remote sensing image processing," *Appl. Soft Comput.*, vol. 64, pp. 75–93, Mar. 2018.

[15] K. C. Mertens, L. P. C. Verbeke, E. I. Ducheyne, and R. R. De Wulf, "Using genetic algorithms in sub-pixel mapping," *Int. J. Remote Sens.*, vol. 24, no. 21, pp. 4241–4247, 2003.

[16] Y. Zhong and L. Zhang, "Remote sensing image subpixel mapping based on adaptive differential evolution," *IEEE Trans. Syst., Man, Cybern. B. Cybern.*, vol. 42, no. 5, pp. 1306–1329, Oct. 2012.

[17] Y. Zhong and L. Zhang, "Sub-pixel mapping based on artificial immune systems for remote sensing imagery," *Pattern Recognit.*, vol. 46, no. 11, pp. 2902–2926, 2013.

[18] Q. Wang, L. Wang, and D. Liu, "Particle swarm optimization-based sub-pixel mapping for remote-sensing imagery," *Int. J. Remote Sens.*, vol. 33, no. 20, pp. 6480–6496, 2012.

[19] X. Xu, Y. Zhong, and L. Zhang, "Adaptive subpixel mapping based on a multiagent system for remote-sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 2, pp. 787–804, Feb. 2014.

[20] Y. Ge, S. Li, and V. C. Lakhan, "Development and testing of a subpixel mapping algorithm," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 7, pp. 2155–2164, Jul. 2009.

[21] Y. Ge, Y. Chen, S. Li, and Y. Jiang, "Vectorial boundary-based sub-pixel mapping method for remote-sensing imagery," *Int. J. Remote Sens.*, vol. 35, no. 5, pp. 1756–1768, 2014.

[22] Q. Wang, "Downscaling MODIS images with area-to-point regression Kriging," *Remote Sens. Environ.*, vol. 166, pp. 191–204, Sep. 2015.

[23] Q. Wang, W. Shi, P. M. Atkinson, and E. Pardo-Iguzquiza, "A new geostatistical solution to remote sensing image downscaling," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 386–396, Jan. 2016.

[24] Y. Ge, "Sub-pixel land-cover mapping with improved fraction images upon multiple-point simulation," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 22, no. 1, pp. 115–126, 2013.

[25] J. P. Ardila, V. A. Tolpekin, W. Bijker, and A. Stein, "Markov-random-field-based super-resolution mapping for identification of urban trees in VHR images," *ISPRS J. Photogramm. Remote Sens.*, vol. 66, pp. 762–775, Nov. 2011.

[26] J. Zhao, Y. Zhong, Y. Wu, L. Zhang, and H. Shu, "Sub-pixel mapping based on conditional random fields for hyperspectral remote sensing imagery," *IEEE J. Sel. Topics Signal Process.*, vol. 9, no. 6, pp. 1049–1060, Sep. 2015.

[27] Y. Chen, Y. Ge, G. B. M. Heuvelink, J. Hu, and Y. Jiang, "Hybrid constraints of pure and mixed pixels for soft-then-hard super-resolution mapping with multiple shifted images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 5, pp. 2040–2052, May 2015.

[28] Y. Zhong, Y. Wu, X. Xu, and L. Zhang, "An adaptive subpixel mapping method based on MAP model and class determination strategy for hyperspectral remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1411–1426, Mar. 2015.

[29] F. Ling, X. Li, F. Xiao, and Y. Du, "Superresolution land cover mapping using spatial regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 7, pp. 4424–4439, Jul. 2014.

[30] R. Feng, Y. Zhong, Y. Wu, D. He, X. Xu, and L. Zhang, "Nonlocal total variation subpixel mapping for hyperspectral remote sensing imagery," *Remote Sens.*, vol. 8, p. 250, Mar. 2016.

[31] X. Xu, Y. Zhong, L. Zhang, and H. Zhang, "Sub-pixel mapping based on a map model with multiple shifted hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 580–593, Apr. 2013.

[32] R. Feng, Y. Zhong, X. Xu, and L. Zhang, "Adaptive sparse subpixel mapping with a total variation model for remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 5, pp. 2855–2872, May 2016.

[33] F. Ling, W. Li, Y. Du, and X. Li, "Land cover change mapping at the subpixel scale with different spatial-resolution remotely sensed imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 1, pp. 182–186, Jan. 2011.

[34] X. Li, Y. Du, and F. Ling, "Super-resolution mapping of forests with bitemporal different spatial resolution images based on the spatial-temporal Markov random field," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 1, pp. 29–39, Jan. 2014.

[35] X. Li, F. Ling, G. M. Foody, Y. Ge, Y. Zhang, and Y. Du, "Generating a series of fine spatial and temporal resolution land cover maps by fusing coarse spatial resolution remotely sensed images and fine spatial resolution land cover maps," *Remote Sens. Environ.*, vol. 196, pp. 293–311, Jul. 2017.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:58:24 UTC from IEEE Xplore. Restrictions apply.

[36] Q. Wang, W. Shi, and P. M. Atkinson, "Spatiotemporal subpixel mapping of time-series images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 9, pp. 5397–5411, Sep. 2016.
[37] F. Gao, J. Masek, M. Schwaller, and F. Hall, "On the blending of the Landsat and MODIS surface reflectance: Predicting daily Landsat surface reflectance," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 8, pp. 2207–2218, Aug. 2006.
[38] I. Manakos and M. Braun, *Land Use and Land Cover Mapping in Europe: Practices & Trends*, vols. 11–30. Amsterdam, The Netherlands: Springer, 2014.
[39] Q. Yu *et al.*, "A survey-based exploration of land-system dynamics in an agricultural region of Northeast China," *Agricult. Syst.*, vol. 121, pp. 106–116, Oct. 2013.
[40] P. Fu and Q. Weng, "A time series analysis of urbanization induced land use and land cover change and its impact on land surface temperature with landsat imagery," *Remote Sens. Environ.*, vol. 175, pp. 205–214, Mar. 2016.
[41] X. Chen, D. Wang, J. Chen, C. Wang, and M. Shen, "The mixed pixel effect in land surface phenology: A simulation study," *Remote Sens. Environ.*, vol. 211, pp. 338–344, Jun. 2018.
[42] D. E. Ahl, S. T. Gower, S. N. Burrows, N. V. Shabanov, R. B. Myneni, and Y. Knyazikhin, "Monitoring spring canopy phenology of a deciduous broadleaf forest using MODIS," *Remote Sens. Environ.*, vol. 104, no. 1, pp. 88–95, 2006.
[43] M. Hussain, D. Chen, A. Cheng, H. Wei, and D. Stanley, "Change detection from remotely sensed images: From pixel-based to object-based approaches," *ISPRS J. Photogramm. Remote Sens.*, vol. 80, pp. 91–106, Jun. 2013.
[44] A. Kanemura, S.-I. Maeda, and S. Ishii, "Superresolution with compound Markov random fields via the variational EM algorithm," *Neural Netw.*, vol. 22, no. 7, pp. 1025–1034, 2009.
[45] L. Zhang, H. Zhang, H. Shen, and P. Li, "A super-resolution reconstruction algorithm for surveillance images," *Signal Process.*, vol. 90, no. 3, pp. 848–859, 2010.
[46] Y.-R. Li, D.-Q. Dai, and L. Shen, "Multiframe super-resolution reconstruction using sparse directional regularization," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 20, no. 7, pp. 945–956, Jul. 2010.
[47] M. K. Ng, H. Shen, E. Y. Lam, and L. Zhang, "A total variation regularization based super-resolution reconstruction algorithm for digital video," *EURASIP J. Adv. Signal Process.*, vol. 2007, p. 074585, Dec. 2007.
[48] D. He, Y. Zhong, R. Feng, and L. Zhang, "Spatial-temporal sub-pixel mapping based on swarm intelligence theory," *Remote Sens.*, vol. 8, no. 12, p. 894, 2016.
[49] D. He, Y. Zhong, and L. Zhang, "Spatiotemporal subpixel geographical evolution mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2198–2220, Apr. 2019.
[50] Q. Wang and P. M. Atkinson, "The effect of the point spread function on sub-pixel mapping," *Remote Sens. Environ.*, vol. 193, pp. 127–137, May 2017.



Da He (M'16) received the B.S. degree in remote sensing science and technology from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China, in June 2015, where he is currently pursuing the Ph.D. degree in photogrammetry and remote sensing with the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing.

His research interests include multi and hyperspectral remote sensing image processing, sub-pixel mapping, space–time analysis, change detection, deep learning, and geographic application.



Yanfei Zhong (M'11–SM'15) received the B.S. degree in information engineering and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2002 and 2007, respectively.

Since 2010, he has been a Full professor with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University. He organized the Intelligent Data Extraction, Analysis and Applications of Remote Sensing (RSIDEA) Research Group. He has published more than 100 research articles in international journals, such as *Remote Sensing of Environment*, *ISPRS Journal of Photogrammetry and Remote Sensing*, and the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. His research interests include hyperspectral remote sensing information processing, high-resolution remote sensing image understanding, and geoscience interpretation for multisource remote sensing data and applications.

Dr. Zhong is a fellow of the Institution of Engineering and Technology (IET). He was a recipient of the 2016 Best Paper Theoretical Innovation Award from the International Society for Optics and Photonics (SPIE). He won the Second-Place Prize in 2013 IEEE GRSS Data Fusion Contest and the Single-view Semantic 3-D Challenge of the 2019 IEEE GRSS Data Fusion Contest, respectively. He is currently serving as an Associate Editor for the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, and the *International Journal of Remote Sensing*.



Liangpei Zhang (M'06–SM'08–F'19) received the B.S. degree in physics from Hunan Normal University, Changsha, China, in 1982, the M.S. degree in optics from the Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, China, in 1988, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 1998.

From 2011 to 2016, he was a Principal Scientist for the China State Key Basic Research Project appointed by the Ministry of National Science and Technology of China to lead the Remote Sensing Program in China. He is currently a Chang-Jiang Scholar Chair Professor appointed by the Ministry of Education, China, State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing (LIESMARS), Wuhan University. He has published more than 700 research articles and five books. He is also the Institute for Scientific Information (ISI) highly cited author. He holds 15 patents. His research interests include hyperspectral remote sensing, high-resolution remote sensing, image processing, and artificial intelligence.

Dr. Zhang is a fellow of the Institution of Engineering and Technology (IET). He was a recipient of the 2010 Best Paper Boeing Award, the 2013 Best Paper ERDAS Award from the American Society of Photogrammetry and Remote Sensing (ASPRS) and 2016 Best Paper Theoretical Innovation Award from the International Society for Optics and Photonics (SPIE). His research teams won the top three prizes of the IEEE GRSS 2014 Data Fusion Contest, and his students have been selected as the winners or finalists of the IEEE International Geoscience and Remote Sensing Symposium (IGARSS) student paper contest in recent years. He also serves as an Associate Editor or Editor of more than ten international journals. He is currently serving as an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. He is also the Founding Chair of the IEEE Geoscience and Remote Sensing Society (GRSS) Wuhan Chapter.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:58:24 UTC from IEEE Xplore. Restrictions apply.